

#2 Free Electron Model of Metals

Drude model

Basic assumptions

- (a) Free electron approximation.
- (b) Electrical resistance results from scattering (all information is lost upon scattering).
- (c) Relaxation-time approximation. (d) Thermal equilibrium.

With(out) electric field Free electron moving randomly, no net current; Or the drift velocity $v_d \ll v_{\text{rms}}$.

EOM

- (a) Let τ be the scattering time and $1/\tau$ be the scattering rate.
- (b) In dt , a fraction $\frac{dt}{\tau}$ of the electrons are scattered with zero average velocity.
- (c) The rest part $1 - dt/\tau$ are accelerated by external force $-eE$.
- (d) In $t \sim t + dt$, the EOM (omit high-order terms)

$$m \, dv/dt = -m\mathbf{v}/\tau - e\mathbf{E}.$$

At steady state ($\frac{dv}{dt} = 0$). The drift velocity

$$\mathbf{v}_d \equiv \langle \mathbf{v}(t) \rangle = -e\tau \mathbf{E}/m \equiv -\mu \mathbf{E},$$

where mobility $\mu = \frac{v}{E} = \frac{e\tau}{m}$. For a given field, a higher μ means the electrons will move faster. The conductance

$$\sigma = \mathbf{j}/\mathbf{E} = -en\mathbf{v}/\mathbf{E} = ne\mu = -e^2 n\tau/m^*.$$

Here, $\tau = l/v_{\text{rms}}$ is usually independent of E .

Classical Hall effect

- (a) Classically, electrons experience $\mathbf{F} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$
- (b) Cyclotron motion: $\mathbf{E} = 0$
 $\ddot{x} + (eB_z/m^*)^2 x = 0, \quad \omega_c = eB/m^*, \quad r_c = v/\omega_c \propto 1/B.$
 Classically, any orbit size is allowed.

Drude model with a magnetic field

Consider Drude model with applied electric and magnetic fields

$$m^* \dot{\mathbf{v}} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - m^* \mathbf{v}/\tau$$

Taking the steady state, with Ohm's rule $\mathbf{j} = -ne\mathbf{v} = \hat{\sigma} \mathbf{E}$

$$\hat{\sigma} = \begin{pmatrix} \frac{\sigma_0}{1+\omega_c^2 \tau^2} & \frac{\sigma_0 \omega_c \tau}{1+\omega_c^2 \tau^2} & 0 \\ -\frac{\sigma_0 \omega_c \tau}{1+\omega_c^2 \tau^2} & \frac{\sigma_0}{1+\omega_c^2 \tau^2} & 0 \\ 0 & 0 & \sigma_0 \end{pmatrix}, \quad \hat{\rho} = \hat{\sigma}^{-1}, \quad \sigma_0 = \frac{ne^2 \tau}{m^*}.$$

The Hall coefficient $R_H = \rho_{xy}/B = -1/ne$.

Drawbacks

- (a) Existence of insulators. (b) Temperature dependence of σ .
- (c) Heat capacity $C_v^D = \frac{3}{2}nk_B$ does not match $C_v \approx 10^{-4}nk_B T$.

Summerfeld model for electrons in metals

- (a) Electrons treated quantum mechanically: Quantize the energy of electron gas.
- (b) The electronic velocity distribution is given by the quantum Fermi-Dirac distribution.
- (c) Pauli exclusion: At most one electron can occupy any single electron level.

Electron wave

Schrödinger equation for free electron $\nabla^2 \psi(\mathbf{r}) = -\frac{2mE}{\hbar^2} \psi(\mathbf{r})$. The solution is a plane wave with the energy

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i(k_x x + k_y y + k_z z)}, \quad E = \frac{\hbar^2(k_x^2 + k_y^2 + k_z^2)}{2m}.$$

Born-von Karman periodic boundary conditions

The electrons are confined in a 3D box ($V = L_x L_y L_z$) with

- (a) $U(\mathbf{r}) = 0$ inside the box, (b) $U(\mathbf{r}) = \infty$ outside the box.

The PBC can be applied

$\psi(x + L_x, y, z) = \psi(x, y + L_y, z) = \psi(x, y, z + L_z) = \psi(x, y, z)$, determines the allowed k

$$e^{ik_i L_i} = 1, \quad k_i = 2\pi n_i / L_i, \quad n_i \in \mathbb{Z}.$$

The boundary conditions lead to quantization.

States in k -space

- (a) The allowed quantum states can be visualized as a 3D grid of points in k -space with the distance of $\Delta k_i = 2\pi/L_i$.
- (b) In k -space, one grid point occupies $\prod_{x,y,z} (\frac{2\pi}{L_i}) = (2\pi)^3/V$.
- (c) There are $2 \times V/(2\pi)^3$ (Each grid point can be occupied by $|\uparrow\rangle, |\downarrow\rangle$) quantum states per unit volume in k -space.

Fermi energy

- (a) The free electron gas is isotropic, so the surface of constant $E = \hbar^2(k_x^2 + k_y^2 + k_z^2)/2m$ in k -space is a sphere.
- (b) Fermi energy: energy of the highest occupied state at zero temperature.
 - i. Number of electrons: $N = 2 \times \frac{4\pi k_F^3/3}{(2\pi/L)^3} = k_F^3 V / 3\pi^2$.
 - ii. Fermi wave vector: $k_F = (3\pi^2 n)^{1/3}$, $n = N/V$ denotes the carrier density.
 - iii. Fermi energy: $E_f = \hbar^2 k_F^2 / 2m = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}$.
 - iv. Fermi velocity and temperature relate to the Fermi energy
 $E_F = \frac{1}{2} m v_F^2 = k_B T_F$.

Bosons and Fermions at $T = 0$

- (a) Bosons: The occupation number can take on any value: photons, mesons, any composite particles with integer spin.
- (b) Fermions: The occupation number can be zero or one due to the Pauli exclusion principle: electrons, protons, neutrons, any composite particles with half-integer spin.

Bosons and Fermions at $T \neq 0$

At finite temperature, Bosons/Fermions obeys Bose-Einstein/Fermi-Dirac statistics $\bar{n}_i = g_i \{ \exp[(\epsilon_i - \mu)/k_B T] \mp 1 \}^{-1}$.

- (a) ϵ_i : Energy of the i -th state. (c) g_i : Degeneracy.
- (b) n_i : Occupation number. (d) μ : Chemical potential.

Bosons and Fermions are identical and indistinguishable particles

The electron gas at zero temperature

- (a) All quantum states inside the Fermi sphere are filled while outside are empty.
- (b) The surface of the Fermi sphere is called Fermi surface.

Fermi-Dirac distribution

Probability distribution function $f(E)$: the probability that a state with energy E is occupied.

$$f(E) = [1 + e^{(E-E_f)/k_B T}]^{-1} \xrightarrow{T \rightarrow 0} \begin{cases} 0, & E > E_f, \\ 1, & E \leq E_f. \end{cases}$$

As T increasing, the function will broaden $\sim 2k_B T$ at $E = E_f$.

- (a) At Fermi level, the probability being occupied is always $1/2$.
- (b) $E_f(T = 0) = E_F$, $E_f(T > 0) < E_F$.

Density of states The number states can be occupied per energy.

$$g(E) = \frac{dn}{dE} = \frac{1}{V} \frac{dN}{dE} = \begin{cases} \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^* E}, & 3D, \quad m^*/\pi \hbar^2, \quad 2D, \\ \frac{1}{\pi \hbar} \sqrt{2m^*/E}, & 1D, \quad 2\delta(E), \quad 0D. \end{cases}$$

The electron gas at $T > 0$

(a) The total number of electron at $T > 0$

$$n = N/V = \int_0^\infty g(E) [e^{(E-E_f)/k_B T} + 1]^{-1} dE.$$

(b) The energy density $u = U/V = \int_0^\infty E g(E) \frac{1}{e^{(E-E_f)/k_B T} + 1} dE$.

Current density

$$\mathbf{j} = -nev = -2e \int \frac{d^3 \mathbf{k}}{(2\pi)^3} f(\mathbf{k}) v(\mathbf{k}).$$

At zero field, for every occupied state $\mathbf{k}$, there is a state $-\mathbf{k}$ occupied with the same probability. Therefore, $\mathbf{j} = 0$.

In Drude model, assume that scattering adds damping

$$\frac{d\hbar \mathbf{k}(t)}{dt} = -e\mathbf{E} - \frac{\hbar[\mathbf{k}(t) - \mathbf{k}]}{\tau} \xrightarrow[t=\infty]{\text{steady state}} \mathbf{k}(t) = \mathbf{k} - \frac{e\tau}{\hbar} \mathbf{E},$$

the wave vector of every electron is shifted. The current density

$$\mathbf{j} = -2e \int \frac{d^3 \mathbf{k}}{(2\pi)^3} f\left(\mathbf{k} + \frac{e\tau}{\hbar} \mathbf{E}\right) v(\mathbf{k}) \frac{\int d^3 \mathbf{k} f(\mathbf{k}) \mathbf{k} = 0}{m} \frac{e^2 \tau}{m} n \mathbf{E}$$

with $\mathbf{j} = \sigma \mathbf{E}$, $\sigma = ne^2 \tau / m$ the same as that in Drude model.

The excess current carriers are only those very near the Fermi surface. So the current carriers have velocity v_F .

Heat capacity of an electron gas At low temperature, $\mu \sim E_f$ and $\frac{\partial f(E,T)}{\partial T}$ is peaked close to E_f , $k_B T \ll E_f$. Then

$$C_e = \partial \Delta U / \partial T \approx \pi^2 n k_B^2 T / 2T_f \propto T.$$

The heat capacity of electrons increases linearly with temperature.

Thermal transport Wiedemann-Franz law: Lorentz number

$$L = \kappa / T \sigma = \frac{\pi^2}{3} \frac{k_B^2}{e^2}.$$

Limitation of Free electron gas model

- (a) Cannot explain semiconductors/insulators.
- (b) $C_V \propto T$ only at very low T .
- (c) Fail to explain $C_V \sim T^3$ at intermediate temperature region.
- (d) Wiedemann-Franz law temperature dependent.
- (e) The sign of Hall coefficient $R_H = -1/ne$.

#3 Lattice Vibrations and Phonons

Specific heat capacity in solids Experimentally, $C_V \sim aT + bT^3$ (electronic and lattice). At high T , C_V approaches constant.

Dulong-Petit law Each atom has 3 independent harmonic oscillators; each contributes $k_B T$ to energy. Thus, $C_V = 3N_A k_B = 3R$.

- (a) Only applicable to heavier elements (fail for light and strongly bonded atoms, such as diamond)
- (b) Applicable only at high temperatures.

Einstein model Assume independent harmonic oscillators with Einstein frequency ω_E and quantized energy $E_n = \hbar \omega_E \left(n + \frac{1}{2}\right)$. N vibrating atoms need $3N$ independent 1D oscillators to describe the motion. The heat capacity can be obtained

$$C = \frac{\partial \bar{E}}{\partial T} = - \frac{\partial}{\partial T} \left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right) = k_B (\beta \hbar \omega_E)^2 \frac{e^{\beta \hbar \omega_E}}{(e^{\beta \hbar \omega_E} - 1)^2}$$

- (a) High temperature ($\beta \rightarrow 0$), $C \rightarrow 3Nk_B$, approach to Dulong-Petit law at high temperature.
- (b) Low temperature ($\beta \rightarrow \infty$), $C \rightarrow 0$, but exponentially not T^3 .

Limitation of Einstein model

- (a) Assume all the vibrations share the same frequency.

- (b) At low temperature, $\omega > k_B T / \hbar$ is frozen. The acoustic branch of the vibrations contribute a lot to the heat capacity.

Debye model

- (a) Atoms are coupled to each other, there vibrations generate sound waves with a linear dispersion $\omega(k) = v_s k$.
- (b) The sound waves are quantized similar to light waves, but there are three modes (one longitudinal two transverse) for each $\mathbf{k}$.
- (c) There is a maximum oscillation frequency (Debye frequency), Debye temperature, and Debye radius

$$\omega_D = (6\pi^2 n)^{1/3} v_s, \quad T_D = \hbar \omega_D / k_B, \quad q_D = \omega_D / v_s.$$

The heat capacity

$$C = \frac{\partial \bar{E}}{\partial T} = 9Nk_B \left(\frac{T}{T_D} \right)^3 \int_0^{T_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx.$$

- i. $T \gg T_D$: $C \rightarrow 3Nk_B$.
- ii. $T \ll T_D$: $C \rightarrow \frac{12}{5} \pi^4 N k_B T^3 / T_D^3$.

Limitation of Debye model

- (a) The cutoff seem like a cheat rather than physics.
- (b) The assumption that sound waves exist even for very large values of k , but sound is based on a long-wavelength idea.
- (c) T_D is varying with temperature experimentally.
- (d) Metal's heat capacity also have a term proportional to T .

Vibrations in a monatomic chain Atoms of mass m , spacing a , nearest-neighbor spring constant β . EOM

$$m \ddot{u}_n = \beta(u_{n+1} + u_{n-1} - 2u_n).$$

- (a) Wave solution gives dispersion $\omega = 2\sqrt{\beta/m} |\sin(ka/2)|$.
- (b) For small k , $\omega \approx a\sqrt{\beta/m} |k|$, looks like sound waves. Implies the velocity of sound (phase velocity) $v_s = v_p = \frac{\omega}{k} = a\sqrt{\frac{\beta}{m}}$.
- (c) At the boundary of 1st BZ, $\omega_{\max} = 2\sqrt{\beta/m}$, $v_g = \frac{\partial \omega}{\partial k} = 0$.

Quantum modes: phonons Refer to a phonon as a Quantized lattice vibration.

- (a) For a normal mode of frequency ω , $E_n = (n + \frac{1}{2}) \hbar \omega$.
- (b) Phonons are bosons, occupation given by Bose-Einstein distribution $n_B(\omega) = [e^{\hbar \omega / k_B T} - 1]^{-1}$.

Crystal momentum Phonon carries crystal momentum $\hbar \mathbf{k}$ (quasi-momentum, defined modulo reciprocal lattice vectors). Conservation of crystal momentum governs scattering processes.

Vibrations in a diatomic chain Two atoms per unit cell (masses m, M) with the dispersion relation

$$\omega_{\pm}^2 = \frac{\beta(M+m)}{Mm} \left[1 \pm \sqrt{1 - \frac{4Mm}{(M+m)^2} \sin^2 ak} \right],$$

gives the acoustic (ω_-) and optical (ω_+) branches.

- (a) $k = 0$, $\omega_-^2 = 0$, $\omega_+^2 = 2\beta/\mu$.
- (b) $k = \pi/(2a)$, $\omega_+^2 = 2\beta/m$, $\omega_-^2 = 2\beta/M$.

The gap at $\pm \frac{\pi}{2a}$ exists if $M \neq m$.

- (a) Acoustic mode: Atoms vibrate in same direction for ω_- .
- (b) Optical mode: Atoms vibrate in opposite direction for ω_+ , can couple to electromagnetic radiation (infrared).

Vibrations in 3D lattices For a unit cell with n atoms, there are $3n$ branches 3 acoustic (one longitudinal, two transverse) and $3n - 3$ optical branches.

Experimental observations of vibrations X-ray scattering, neutron scattering, Raman spectroscopy, Brillouin spectroscopy, etc., used to measure dispersion $\omega(k)$ and density of states $g(\omega)$.

#4 Crystal Symmetry and Structure

Crystal definition Periodic, Basis, Symmetry, Anisotropy.

Bravais lattice A lattice is repeating structure defined as an infinite set of points

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3,$$

which can be constructed as sums of a set of linearly independent primitive vectors $\mathbf{a}_i$. With the properties of *identical*, *infinite extension*, and *contains crystal symmetry*.

Primitive unit cell Contains only one lattice point, with volume $\Omega = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$.

Wigner-Seitz cell Lines perpendicular and connecting the mid-points between a given lattice and all nearby lattice sites.

Crystallographic directions and planes Crystallographic directions $[l_1l_2l_3]$ represents $\mathbf{n} = l_1\mathbf{a}_1 + l_2\mathbf{a}_2 + l_3\mathbf{a}_3$. A group of identical directions can be written as $\langle l_1l_2l_3 \rangle$, like

$$\langle 100 \rangle = \{ [100], [\bar{1}00], [010], [0\bar{1}0], [001], [00\bar{1}] \}.$$

Planes: Miller indices (hkl) obtained from reciprocals of intercepts

$$1/a : 1/b : 1/c = h : k : l$$

with the corresponding family $\{hkl\}$.

Crystal symmetry

Symmetry operations

- (a) n -fold rotation axis: An object that requires rotation of $2\pi/n$
- (b) Reflection (mirror), Inversion, Rotoinversion.

Crystal structures

- CsCl: Simple cubic with basis Cs at $[0, 0, 0]$, Cl at $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$.
- NaCl: FCC with basis Na at $[0, 0, 0]$, Cl at $[\frac{1}{2}, 0, 0]$.
- Diamond/Zinc blende (ZnS): FCC with basis two C (or Zn and S) atoms at $[0, 0, 0]$ and $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$.
- Perovskite (ABO_3): Cubic with A at corners, B at body center, O at face centers.

Reciprocal lattice Defined by $\mathbf{G} \cdot \mathbf{R} = 2\pi m$. Primitive vectors

$$\mathbf{b}_i = \frac{2\pi(\mathbf{a}_j \times \mathbf{a}_k)}{\mathbf{a}_i \cdot (\mathbf{a}_j \times \mathbf{a}_k)}, \quad \mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}, \quad \Omega = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3).$$

(a) BCC Lattice

$$\mathbf{a}_1 = \frac{a}{2}(-1, 1, 1), \quad \mathbf{a}_2 = \frac{a}{2}(1, -1, 1), \quad \mathbf{a}_3 = \frac{a}{2}(1, 1, -1),$$

$$\mathbf{b}_1 = \frac{2\pi}{a}(0, 1, 1), \quad \mathbf{b}_2 = \frac{2\pi}{a}(1, 0, 1), \quad \mathbf{b}_3 = \frac{2\pi}{a}(1, 1, 0).$$

$$\mathbf{G} = \sum_j n_j \mathbf{b}_j = \frac{2\pi}{a}(n_2 + n_3, n_1 + n_3, n_1 + n_2).$$

(b) FCC Lattice

$$\mathbf{a}_1 = \frac{a}{2}(0, 1, 1), \quad \mathbf{a}_2 = \frac{a}{2}(1, 0, 1), \quad \mathbf{a}_3 = \frac{a}{2}(1, 1, 0),$$

$$\mathbf{b}_1 = \frac{2\pi}{a}(-1, 1, 1), \quad \mathbf{b}_2 = \frac{2\pi}{a}(1, -1, 1), \quad \mathbf{b}_3 = \frac{2\pi}{a}(1, 1, -1).$$

$$\mathbf{G} = \frac{2\pi}{a}(-n_1 + n_2 + n_3, n_1 - n_2 + n_3, n_1 + n_2 - n_3).$$

Reciprocal lattice vectors are normal to real lattice planes (hkl) ; interplanar spacing $d_{hkl} = 2\pi/|\mathbf{G}_{hkl}|$. See **Problem 3.3**.

$$d_{h_1h_2h_3} = \frac{2\pi}{|\mathbf{G}_{h_1h_2h_3}|} = \frac{2\pi}{|h_1\mathbf{b}_1 + h_2\mathbf{b}_2 + h_3\mathbf{b}_3|}.$$

(a) For cubic

$$d_{h_1h_2h_3} = a(h_1^2 + h_2^2 + h_3^2)^{1/2}.$$

(b) For tetragonal

$$d_{h_1h_2h_3} = a\left(\frac{h_1^2 + h_2^2}{a^2} + \frac{h_3^2}{c^2}\right)^{1/2}.$$

(c) For hexagonal

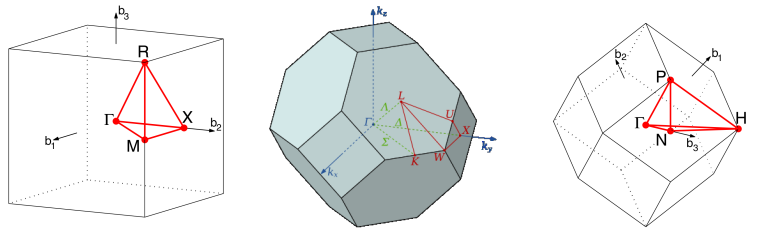
$$d_{h_1h_2h_3} = a\left[\frac{3}{4}(h_1^2 + h_1h_2 + h_2^2) + \left(\frac{a}{c}\right)^2 h_3^2\right]^{1/2}.$$

(d) For orthorhombic

$$d_{h_1h_2h_3} = \left[(h_1/a)^2 + (h_2/b)^2 + (h_3/c)^2\right]^{-1/2}.$$

Brillouin zone Wigner-Seitz unit cell of reciprocal lattice is the first BZ, contains all wavevectors closer to origin than to any other reciprocal lattice point.

- (a) SC. $\Gamma : (0, 0, 0)$, $X : \pi/a(1, 0, 0)$, $R : \pi/a(1, 1, 1)$.
- (b) FCC. $\Gamma : (0, 0, 0)$, $X : 2\pi/a(1, 0, 0)$,
 $L : \pi/a(1, 1, 1)$, $R : 2\pi/a(3/4, 3/4, 1)$.
- (c) BCC. $\Gamma : (0, 0, 0)$, $X : 2\pi/a(1, 0, 0)$,
 $P : \pi/a(1, 1, 1)$, $N : 2\pi/a(1/2, 1/2, 0)$.



Chemical bonding

Cohesive energy = energy of free atoms – crystal energy.

(a) van der Waals (weak, induced dipole)

$$U = 4\epsilon \left[\left(\frac{\sigma}{R} \right)^{12} - \left(\frac{\sigma}{R} \right)^6 \right].$$

- (b) Ionic crystals (electrostatic, strong)
- (c) Covalent (electron sharing, directional)
- (d) Metallic (delocalized electrons)
- (e) Hydrogen bonding (dipole-dipole).

#5 Scattering Experiments

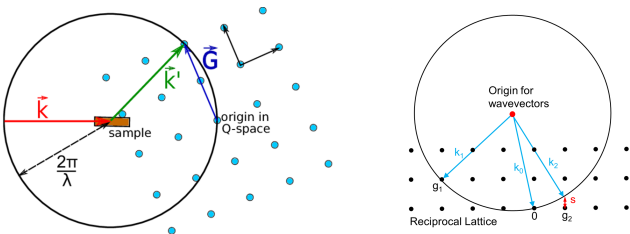
X-Ray

- (a) X-rays are electromagnetic waves. Atoms scatter X-ray waves elastically, primarily through the atoms' electrons.
- (b) Some scattered waves can add constructively (constructive interference) in a few specific directions.
- (c) The diffraction pattern of X-ray is highly relative to the crystal structure.

Bragg's Law $2d_{hkl} \sin \theta = n\lambda$.

Laue's condition Under elastic scattering, $\mathbf{k}' - \mathbf{k} = \mathbf{G}$.

Ewald construction for diffraction



- (a) Draw a sphere in reciprocal space centered on the tip of the incident wave vector $\mathbf{k}$ of radius k (so that it passed through the origin).
- (b) There will be some wave vector satisfying the Laue condition if and only if some reciprocal lattice point (in addition to the origin) lies on the surface of the sphere).

Geometrical structure factor $S_G = \sum_j f_j e^{-i\mathbf{G}\cdot\mathbf{r}_j}$. f is the atomic form factor of the specific atom, the sum is relative to the structure inside a basis.

For BCC Structure, there are two atoms in a unit cell: $(0, 0, 0)$ and $\frac{a}{2}(1, 1, 1)$. Then

$$\mathbf{G} = \frac{2\pi}{a}(h\hat{x} + k\hat{y} + l\hat{z}), \quad S_G = f(1 + e^{-i\pi(h+k+l)}).$$

- (a) $S_G = 0$, when $h + k + l$ is odd. No diffraction.
- (b) $S_G = 2f$, when $h + k + l$ is even. Diffraction peak exists.

For BCC Structure, there are four atoms in a unit cell: $(0, 0, 0)$, $\frac{a}{2}(1, 1, 0)$, $\frac{a}{2}(0, 1, 1)$, and $\frac{a}{2}(1, 0, 1)$. Then

$$S_G = f(1 + e^{-i\pi(h+k)} + e^{-i\pi(k+l)} + e^{-i\pi(h+l)}).$$

- (a) $S_G = 2f$, when h, k, l are all even or odd. Diffraction peak exists.
- (b) $S_G = 0$, otherwise. No diffraction.

Diffraction methods Laue method, Rotating crystal method, and Powder diffraction. See **Problem 4.2**.

Electron diffraction

- (a) Electrons interact strongly with matter, small penetration.
- (b) Wavelength $\lambda = \sqrt{1.5/V}$ nm (V in kV).
- (c) Used in transmission electron microscopy (TEM).

Neutron diffraction

- (a) Neutrons interact with nuclei and magnetic moments.
- (b) Penetrating, sensitive to light elements and magnetic structure.
- (c) Wavelength $\sim 1 \text{ \AA}$, energy $\sim 0.08 \text{ eV}$.

#6 Bloch Theorem

For electron in periodic potential $U(\mathbf{r} + \mathbf{R}) = U(\mathbf{r})$, the wavefunction has form

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), \quad u_{\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{\mathbf{k}}(\mathbf{r}).$$

- (a) Crystal momentum $\hbar\mathbf{k}$.
- (b) Energy bands $E_n(\mathbf{k})$ periodic in $\mathbf{k}$ with period $\mathbf{G}$.

Properties of Bloch function

- (a) The wavefunction of electron in a crystal is the combination of free electron and isolated atom.
- (b) If the electron is fully free, $u_{\mathbf{k}}(\mathbf{r}) = A = \text{Const.}$
- (c) If the electron is fully bound, $e^{i\mathbf{k}\cdot\mathbf{r}} = \text{Const.}$
- (d) The periodic function $u_{\mathbf{k}}(\mathbf{r})$ reflects the interaction of electron and the crystal.

$$\begin{cases} \psi_{\mathbf{k}}(\mathbf{r}) = A e^{i\mathbf{k}\cdot\mathbf{r}}, & \text{Free electron,} \\ \psi(\mathbf{r}) = C u(\mathbf{r}), & \text{Isolated atom,} \\ \psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}), & \text{Electron in a crystal} \end{cases}$$

Periodicity of Bloch's theorem

- (a) The potential can be in a Fourier series $U(\mathbf{r}) = \sum_{\mathbf{G}} U_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}$.
- (b) The allowed vectors of $\mathbf{G}$ are given by $e^{i\mathbf{G}\cdot\mathbf{R}} = 1$.
- (c) If the Bloch form holds for $\mathbf{k}' = \mathbf{k} + \mathbf{G}$, it will also hold for $\mathbf{k}$.

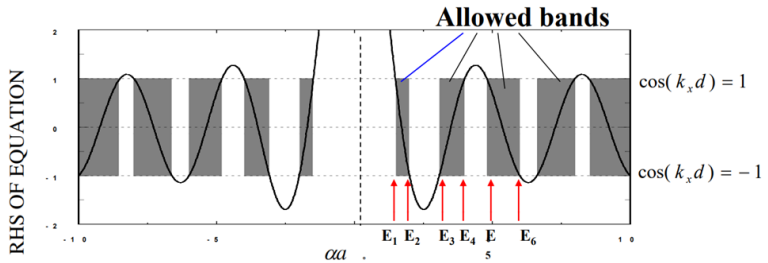
- (d) The wave vector can be restricted to the first Brillouin zone of the reciprocal lattice without loss of generality.
- (e) The boundary conditions make k discrete.

Brillouin zone Wavevector $\mathbf{k}$ can be restricted to first Brillouin zone. Number of allowed $\mathbf{k}$ -states equals number of unit cells.

Kronig-Penney model 1D periodic square-well potential. Dispersion given by

$$\cos(kd) = \cos(\alpha a) \text{ch}(\beta b) - \frac{\alpha^2 - (\beta)^2}{2i\alpha\beta} \sin(\alpha a) \text{sh}(\beta b),$$

where $d = a(\text{Width of } V = 0) + b(\text{Width of } V = V_0)$, $\alpha^2 = 2mE/\hbar^2$, $\beta^2 = 2m(V_0 - E)/\hbar^2$.



Solutions exist only for certain energy ranges (bands), separated by gaps due to $\cos(kd) \in [-1, 1]$.

Crystal momentum (quasi-momentum) It is conserved. For a photon to change the crystal momentum of an electron in the reduced zone scheme requires the involvement of a phonon.

#7 Band Theory I-NFE and TB model

Nearly free electron (NFE) model

- (a) Weak periodic potential treated as perturbation.
- (b) Unperturbed states are plane waves.
- (c) The wave function of unperturbed electrons can be expressed by a plane wave $|\mathbf{k}\rangle$.
- (d) When the Bragg condition $\mathbf{k}' = \mathbf{k} \pm \mathbf{G}$ satisfies, the electron wave is reflected and changed into a reflected wave state $|\mathbf{k}'\rangle$.
- (e) The combination of the incident plane wave and the reflected plane wave leads to the standing wave and energy gap.

For 1D Degenerate case, the dispersion relation

$$E_{\pm} = \frac{1}{2} \left\{ E_k^{(0)} + E_{k'}^{(0)} \pm \sqrt{[E_{k'}^{(0)} - E_k^{(0)}]^2 + 4|U_n|^2} \right\}.$$

Taking $|E_{k'}^{(0)} - E_k^{(0)}| \ll |U_n|$, then the gap magnitude at the boundary of Brillouin zone $k = \pm\pi n/a$ is $E_g = 2|U_n|$, where U_n is Fourier component of potential.

Tight binding (TB) model

- (a) Electrons spend most the time tightly bound to atoms.
- (b) Wavefunction is a linear combination of atomic orbitals

$$\psi(\mathbf{r}) = \sum_m c_m \varphi_i(\mathbf{r} - \mathbf{R}_m).$$

- (c) The total potential in the crystal is the sum of all atomic potentials and has periodicity

$$U(\mathbf{r}) = \sum_m V(\mathbf{r} - \mathbf{R}_m) = U(\mathbf{r} + \mathbf{R}_n).$$

Energy bands

$$E(\mathbf{k}) = \varepsilon_i - \sum_s J(\mathbf{R}_s) e^{-i\mathbf{k}\cdot\mathbf{R}_s},$$

where $J(\mathbf{R}_s)$ is hopping integral. Nearest-neighbor approximation

$$E(\mathbf{k}) = \epsilon_i - J_0 - \sum_{s=\text{NN}} J(\mathbf{R}_s) e^{-i\mathbf{k}\cdot\mathbf{R}_s}$$

is used. For Simple cube, $\mathbf{R}_s = (\pm a, 0, 0), (0, \pm a, 0), (0, 0, \pm a)$.

$$E(\mathbf{k}) = \varepsilon_s - J_0 - 2J_1(\cos k_x a + \cos k_y a + \cos k_z a).$$

In the first BZ of simple cube, there are several high symmetric points $\Gamma = (0, 0, 0)$, $M = (\pi/a, \pi/a, 0)$, $R = (\pi/a, \pi/a, \pi/a)$, $X = (\pi/a, 0, 0)$. Then $\Delta E_{\max} = 12J_1$. Near the band bottom

$$E(\mathbf{k}) \approx \varepsilon_s - J_0 - 2J_1(3 - a^2 k^2/2).$$

The effective mass $m^* = \hbar^2/(2J_1 a^2)$.

Comparison of NFE and TBM

- (a) Nearly free electron model: Good for nearly free electrons (conduction bands).
- (b) TB: Good for tightly bound electrons (valence bands).

#8 Band Theory II

Fermi surface

- (a) Surface in $\mathbf{k}$ -space separating occupied and unoccupied states at $T = 0$.
- (b) Exists only in metals.
- (c) Shape depends on band structure; volume depends on electron density.

Effective mass

- (a) Defined by $m^* = \hbar^2/(\partial^2 E/\partial k^2)$.
- (b) Near band extrema

$$E(\mathbf{k}) \approx E_0 + \frac{\hbar^2}{2m^*}(\mathbf{k} - \mathbf{k}_0)^2.$$

- (c) Can be anisotropic (tensor) and even negative (holes).

Electron and hole Missing electron in valence band acts as positive charge carrier with wavevector $\mathbf{k}_h = -\mathbf{k}_e$ and same velocity as the missing electron.

Metal, insulator, semiconductor

- (a) Metal: partially filled band.
- (b) Insulator: large band gap, filled valence band.
- (c) Semiconductor: small band gap, conductivity increases with temperature.

Direct vs indirect band gap

- (a) Direct: valence band maximum and conduction band minimum at same $\mathbf{k}$.
- (b) Indirect: at different $\mathbf{k}$; optical transitions require phonon assistance.

Optical properties Photon absorption possible if $h\nu \geq E_g$.

- (a) Direct gap materials have strong absorption.
- (b) Indirect gap materials have weaker, phonon-assisted absorption.

Excitons Bound electron-hole pair due to Coulomb attraction. Energy levels below band gap: $E_n = E_g - R_{ex}/n^2$, where R_{ex} is exciton Rydberg.

#9 Physics of Semiconductors

Intrinsic semiconductors For Pure semiconductor, carrier concentrations $n = p = n_i$. At finite T ,

$$n = N_c e^{-(E_c - E_f)/k_B T}, \quad p = N_v e^{-(E_f - E_v)/k_B T},$$

where N_c, N_v are effective densities of states. Intrinsic carrier concentration

$$n_i = \sqrt{N_c N_v} e^{-E_g/(2k_B T)}.$$

Fermi level near midgap: $E_f = E_v + \frac{1}{2}E_g - \frac{3}{4}k_B T \ln(m_e^*/m_h^*)$.

Doped semiconductors

- (a) n-type: donors (group V) add electrons. $n \approx N_d, p = n_i^2/N_d$.
- (b) p-type: acceptors (group III) add holes. $p \approx N_a, n = n_i^2/N_a$.

Fermi level shifts toward donor/acceptor levels.

Carrier scattering

- (a) Impurity scattering: mobility $\propto T^{3/2}/N_i$.
- (b) Lattice scattering: mobility $\propto T^{-3/2}$ (acoustic phonons) or $T^{-1/2}$ (optical phonons).
- (c) Total mobility: $1/\mu_{\text{total}} = \sum 1/\mu_i$.

p-n junction Formed by joining p- and n-type semiconductors.

- (a) Built-in potential $V_0 = \frac{k_B T}{e} \ln(N_a N_d/n_i^2)$.
- (b) Depletion width $W = \sqrt{[2\varepsilon(N_a + N_d)V_0]/(eN_a N_d)}$.
- (c) Current-voltage characteristic (ideal diode)

$$I = I_0(e^{eV/(k_B T)} - 1).$$

- (d) Reverse breakdown: avalanche (impact ionization) or Zener (tunneling).

Devices

- (a) LED: forward-biased p-n junction; recombination emits light.
- (b) Solar cell: photon absorption creates electron-hole pairs, separated by built-in field.
- (c) BJT: npn or pnp structure; amplifies current.
- (d) MOSFET: metal-oxide-semiconductor field-effect transistor; voltage-controlled channel.
- (e) Schottky barrier: metal-semiconductor contact; rectifying or ohmic.

Heterojunctions

- (a) Different semiconductors joined.
- (b) Band alignments: type I (straddling), type II (staggered), type III (broken gap).
- (c) Used in 2D electron gases, optical devices, tunneling diodes.

#10 Electron in a Magnetic Field

Bohr-Sommerfeld quantization

Quantization rule

Number of wavelengths along a closed trajectory must be integer

$$\frac{1}{\hbar} \oint \mathbf{p} \cdot d\mathbf{r} = 2\pi N, \quad N \in \mathbb{Z}.$$

In a magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$, the canonical momentum becomes $\mathbf{p} \rightarrow \mathbf{p} - e\mathbf{A}$. The quantization condition becomes

$$\frac{1}{\hbar} \oint (\mathbf{p} - e\mathbf{A}) \cdot d\mathbf{r} = 2\pi N.$$

This leads to quantization of the magnetic flux enclosed by the orbit

$$\phi = N\phi_0, \quad \phi_0 = \frac{h}{e}.$$

Onsager quantization condition

- (a) In k -space, the area enclosed by an orbit is quantized

$$S_k = \frac{2\pi e B}{\hbar} \left(N + \frac{1}{2} - \frac{\gamma}{2\pi} \right),$$

where γ is the Berry phase.

- (b) For free electrons ($S_k = \pi k^2$), the energy becomes quantized

$$E_N = \hbar\omega_c \left(N + \frac{1}{2} - \frac{\gamma}{2\pi} \right), \quad \omega_c = \frac{eB}{m^*}.$$
- (c) Without Berry phase ($\gamma = 0$), $E_N = \hbar\omega_c(N + 1/2)$.

Landau quantization

Quantum mechanical solution

- (a) Hamiltonian for an electron in a magnetic field (Landau gauge $\mathbf{A} = (0, Bx, 0)$) $\hat{H} = \frac{1}{2m^*}(\mathbf{p} - e\mathbf{A})^2$.
- (b) The motion parallel to $\mathbf{B}$ is free, while the perpendicular motion is quantized into Landau levels (LLs)

$$E = \frac{\hbar^2 k_z^2}{2m^*} + \left(N + \frac{1}{2} \right) \hbar\omega_c, \quad N = 0, 1, 2, \dots$$

- (c) In 2D ($k_z = 0$): $E_N = (N + 1/2)\hbar\omega_c$.

Degeneracy of Landau levels

- (a) Each Landau level has a degeneracy $n_B = g_s eB/h = g_s B/\phi_0$.
- (b) There is one state (or 2 if considering spin) per flux quantum in each Landau level.
- (c) The filling factor $\nu = \frac{n_e}{n_B} = \frac{\hbar n_e}{eB}$ tells how many Landau levels are filled at a given B .

Density of states

- (a) In 3D, the DOS becomes a series of peaks (Landau tubes) broadened by scattering.
- (b) In 2D, the DOS is a set of delta functions at each Landau level, broadened by disorder.
- (c) The broadening $\Delta E \sim \hbar/\tau_q$, where τ_q is the quantum lifetime.
- (d) Landau levels are resolved when $\omega_c\tau_q \gg 1$.

Zeeman splitting

- (a) Zeeman term adds spin splitting

$$E = \left(N + \frac{1}{2} \right) \hbar\omega_c \pm g^* \mu_B B = \left(N + \frac{1}{2} \right) E_c \pm E_z.$$
 - i. An even integer filling factor means that N Landau levels are fully filled.
 - ii. An odd integer filling factor means one spin direction of Landau level is full, while the other is empty.
- (b) The spin susceptibility $\chi = E_z/E_c \propto g^* m^*$.

Shubnikov-de Haas (SdH) oscillations

Basic phenomenology

- (a) SdH oscillations in longitudinal resistivity, the periodicity is given as $\Delta\left(\frac{1}{B}\right) = \frac{2e}{\hbar n_e}$.
- (b) The oscillation frequency $B_F = \frac{\hbar}{2\pi e} S_k$ gives the extremal cross-sectional area of the Fermi surface.

Lifshitz-Kosevich (LK) formalism

- (a) Amplitude of SdH oscillations

$$\frac{\Delta\rho_{xx}}{\rho_0} = R_T R_D R_S \cos\left[2\pi\left(\frac{B_F}{B} + \frac{1}{2} + \beta \right) \right],$$
 where

$$\begin{cases} R_T = \frac{2\pi^2 k_B T / \hbar\omega_c}{\text{sh}(2\pi^2 k_B T / \hbar\omega_c)}, & \text{thermal damping,} \\ R_D = e^{-\pi/\omega_c\tau_q}, & \text{Dingle factor,} \\ R_S = \cos\left(\frac{\pi g^* m^*}{2m_0} \right), & \text{Spin damping.} \end{cases}$$
- (b) Phase factor $\beta = \gamma/2\pi$ contains the Berry phase γ .

Extracting parameters

- (a) From temperature dependence of the amplitude $\rightarrow$ cyclotron mass m^* .

- (b) From Dingle plot $\rightarrow$ quantum lifetime τ_q .
- (c) From phase $\beta \rightarrow$ Berry phase ($\beta = 1/2$ for Dirac fermions).

Integer quantum Hall effect (IQHE)

Basic observations

- (a) At high magnetic fields ($\omega_c\tau \gg 1$), Hall resistivity ρ_{xy} forms plateaus at quantized values, $\rho_{xy} = \frac{\hbar}{\nu e^2}$, $\nu = 1, 2, 3, \dots$
- (b) Resistivity/conductivity $\rho_{xx}, \sigma_{xx} \rightarrow 0$ within the plateaus.
- (c) Quantized Hall conductance $\sigma_{xy} = \nu \frac{e^2}{h}$.

Application to metrology von Klitzing constant $R_k = h/e^2$.

Explanation: Edge states

- (a) In a finite sample, confinement potential creates edge channels.
- (b) Edge states are chiral (unidirectional) and carry the current.
- (c) Backscattering is suppressed $\rightarrow$ dissipationless transport.

Explanation: Localization and plateaus

- (a) Disorder localizes states in the tails of Landau levels.
- (b) Extended states exist only near the center of each Landau level.
- (c) As the Fermi level moves through localized states, σ_{xx} remains zero $\rightarrow$ Hall plateaus.

Fractional quantum Hall effect (FQHE)

Basic observations

- (a) Occurs in very clean 2DEG at high B and low T .
- (b) Hall plateaus at fractional filling factors $\nu = p/q$.
- (c) ρ_{xx} shows minima at the same fillings.

Composite fermion picture

- (a) Each electron binds an even number $2j$ of magnetic flux quanta $\rightarrow$ composite fermion (CF).
- (b) CFs experience an effective magnetic field $B^* = B - 2j\phi_0 n_e$.
- (c) The FQHE of electrons is the IQHE of CFs.
- (d) Principal Jain sequence: $\nu = \frac{j}{2j\pm 1}$.

Wigner crystal

- (a) At very low density and high B , electrons can crystallize into a Wigner crystal.
- (b) Characterized by vanishing conductivity and pinning mode resonances.

#11, #12 Dirac Band Theory

Band structure of monolayer graphene

See **Problem 9.1**. $E = \epsilon_0 \pm |f(\mathbf{k})|$, $f(\mathbf{k}) = \sum_{i=1}^3 e^{i\mathbf{k}\cdot\delta_i}$.

More general expression Consider specific example of two orbitals per unit cell

$$E_{\pm} = \frac{\epsilon_0 \pm \gamma_0 |f(\mathbf{k})|}{1 \mp s_0 |f(\mathbf{k})|}.$$

- E_+ and E_- are the conduction band and valence band.
- (a) Two bands: conduction (π) and valence (π^*) band.
- (b) No energy gap at the $K, K' = \frac{4\pi}{3\sqrt{3}a}(\pm 1, 0)$ points.
- (c) The reasonable case: $\epsilon_0 = 0, s_0 = 0, E_{\pm} = \pm\gamma_0 |f(\mathbf{k})|$. Then at K, K' points $E_{\pm} = 0$. Indicating that there is no coupling between the A and B sublattices. They support the same quantum states, leading to a degeneracy point.
- (d) K, K' points: 4-fold degeneracy = 2 sublattices + 2 valleys.

Linear expansion near K, K' points

See **Problem 9.2**: Resemble massless particle (photons) $E = cp$, but with 300 times slower speed.

Low-energy effective Hamiltonian

(a) Near K ($\xi = +1$) and K' ($\xi = -1$)

$$H_{\xi}(\mathbf{p}) = v_F \begin{pmatrix} 0 & \xi p_x - i p_y \\ \xi p_x + i p_y & 0 \end{pmatrix} = v_F (\xi \sigma_x p_x + \sigma_y p_y).$$

(b) Dispersion: $E_{\pm}(\mathbf{p}) = \pm v_F |\mathbf{p}|$.

(c) Eigenstates (pseudospinors)

$$\psi_{\pm, \xi} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm \xi e^{i \xi \theta_p} \end{pmatrix}, \quad \theta_p = \arctan(p_y/p_x).$$

Pseudospin degree of freedom Pseudospin is a degree of freedom related to the relative amplitude of Bloch function on the A or B sublattice.

Chirality The orientation of the pseudospin is related to the direction of the momentum, The amplitudes on the A or B sublattice of eigenstate depend on $\theta_q = \arctan p_y/p_x$.

Chirality is reversed in the K' valley.

(a) Conduction band electron: $\boldsymbol{\sigma} \cdot \mathbf{n} = 1$.

(b) Valence band electron: $\boldsymbol{\sigma} \cdot \mathbf{n} = -1$.

Chirality conserved: no backscattering between Dirac cones.

Klein tunneling If a given potential doesn't break A-B symmetry, then it is unable to influence the pseudospin degree of freedom, therefore, be conserved upon scattering.

Klein paradox

(a) Transmission of relativistic particles unimpeded even by highest barriers.

(b) Physical picture: particle/hole conversion.

Berry phase The Berry phase is the same for conduction and valley band. Berry phase around K, K' point $\phi_B = \pm \pi$.

Trigonal warping beyond linearization

Consider the next nearest-neighbor hopping t' .

$$E(\mathbf{q}) \approx 3t' \pm v_F |\mathbf{q}| - \left(\frac{9t'^2 a^2}{4} \pm \frac{3t}{8} a^2 \sin 3 \arctan \frac{q_x}{q_y} \right) \mathbf{q}^2$$

Trigonal warping

(a) t' shifts in energy and breaks electron-hole symmetry.

(b) The second order part depends on the direction in momentum space and has a threefold symmetry.

Density of states

(a) Linear DOS near Dirac point

$$g(E) = \frac{2|E|}{\pi \hbar^2 v_F^2}.$$

(b) Carrier density n and Fermi energy

$$E_F = \hbar v_F \sqrt{\pi |n|}, \quad k_F = \sqrt{\pi |n|}.$$

Band structure of bilayer graphene

Tight-binding model

(a) Bernal (AB) stacking: interlayer coupling $\gamma_1 \approx 0.39$ eV between A2 and B1.

(b) Hamiltonian in basis (A1, B1, A2, B2)

$$H = \begin{pmatrix} 0 & -\gamma_0 f & 0 & 0 \\ -\gamma_0 f^* & 0 & \gamma_1 & 0 \\ 0 & \gamma_1 & 0 & -\gamma_0 f \\ 0 & 0 & -\gamma_0 f^* & 0 \end{pmatrix}.$$

(c) Four bands: two high-energy bands split by $\pm \gamma_1$, and two low-energy bands touching at K .

Low-energy effective Hamiltonian

The effective two-component Hamiltonian

$$H_{2, \xi}(\mathbf{p}) = -\frac{1}{2m^*} \begin{pmatrix} 0 & (\xi p_x - i p_y)^2 \\ (\xi p_x + i p_y)^2 & 0 \end{pmatrix}, \quad m^* = \frac{\gamma_1}{2v_F^2}.$$

(a) Parabolic dispersion: $E_{\pm}(\mathbf{p}) = \pm \frac{p^2}{2m^*}$.

(b) Berry phase $\gamma = 2\pi$.

Trigonal warping

(a) Due to skew interlayer hopping γ_3 , the low-energy dispersion becomes anisotropic.

(b) At very low energies, the Fermi surface splits into four Dirac pockets.

Electrically tunable band gap

(a) A perpendicular electric field breaks inversion symmetry $\rightarrow$ opens a band gap.

(b) Gap size $\Delta \propto$ applied field, up to $\sim \gamma_1$.

Landau levels in graphene

Monolayer graphene

(a) Landau level energies

$$E_N = \pm \hbar \omega_c \sqrt{2N}, \quad \omega_c = \frac{v_F}{l_B}, \quad l_B = \sqrt{\frac{\hbar}{eB}}.$$

(b) Zero-energy Landau level ($N = 0$) is half-filled in neutral graphene.

(c) Fourfold degeneracy (spin $\times$ valley).

Bilayer graphene

(a) Landau level energies

$$E_N = \pm \hbar \omega_c \sqrt{N(N-1)}, \quad \omega_c = \frac{eB}{m^*}.$$

(b) Two zero-energy levels ($N = 0, 1$) $\rightarrow$ eightfold degeneracy.

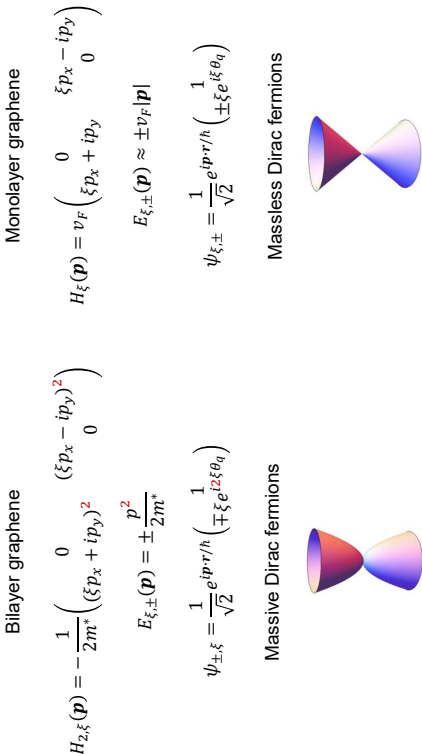
Quantum Hall effect

(a) Monolayer: Hall conductivity $\sigma_{xy} = \pm(4N + 2) \frac{e^2}{h}$ (half-integer QHE).

(b) Bilayer: $\sigma_{xy} = \pm 4N \frac{e^2}{h}$ (integer QHE).

(c) Room-temperature QHE observable due to large Landau level spacing.

Bilayer graphene vs monolayer graphene



Physics in bilayer graphene is quite different from that in monolayer graphene

Homework #1 Free electron model

Problem 1.1 (Two-charge-carrier Drude model). Consider a system of two types of charge carriers in the Drude model. The two carriers have the same density (n) and opposite charge (e and $-e$), and their masses and relaxation rates are m_1, m_2 and τ_1, τ_2 , respectively. So their corresponding mobilities are $\mu_1 = \frac{e\tau_1}{m_1}$ and $\mu_2 = \frac{e\tau_2}{m_2}$, respectively.

- (a) Calculate the magnetoresistance $\Delta\rho = \rho(B) - \rho(B = 0)$, where B is the magnetic field.
- (b) Calculate the Hall coefficient.
- (c) In an undoped semiconductor, $n = n_0 e^{-\Delta/k_B T}$ describes the temperature dependence of the carrier concentration. What will the temperature dependence of the magnetoresistance and the Hall coefficient be?

Solution.

- (a) Assume $\mathbf{B} = (0, 0, B)$. Then, the resistance tensor

$$\hat{\rho} = \hat{\sigma}^{-1} = \frac{1}{\sigma_{xx}^2 + \sigma_{xy}^2} \begin{pmatrix} \sigma_{xx} & -\sigma_{xy} & 0 \\ \sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & (\sigma_{xx}^2 + \sigma_{xy}^2)/\sigma_{zz} \end{pmatrix}$$
$$\sigma_{xx} = \sigma_{yy} = \frac{\sigma_0}{1 + \omega_c^2 \tau_i^2}, \quad \sigma_{xy} = -\sigma_{yx} = \frac{\sigma_0 \omega_c \tau_i}{1 + \omega_c^2 \tau_i^2},$$
$$\sigma_{zz} = \sigma_0 = nq^2 \tau_i / m_i, \quad \omega_c = Bq / m_i,$$

For double charge carriers, $\sigma_{ij}^{\text{tot}} = \sigma_{ij}^{(1)} + \sigma_{ij}^{(2)}$. Then

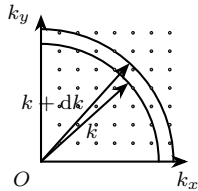
$$\rho_{xx} = \frac{1 + B^2 \mu_1 \mu_2}{ne(\mu_1 + \mu_2)}, \quad \rho_{xy} = \frac{B(\mu_2 - \mu_1)}{ne(\mu_1 + \mu_2)}$$

and the differences

$$\Delta\rho_{xx} = \frac{B^2 \mu_1 \mu_2}{ne(\mu_1 + \mu_2)}, \quad \Delta\rho_{xy} = \frac{B(\mu_1 - \mu_2)}{ne(\mu_1 + \mu_2)}.$$

- (b) $R_H = \frac{\rho_{xy}}{B} = \frac{\mu_2 - \mu_1}{ne(\mu_1 + \mu_2)}$. (c) $R_H, \Delta\rho \propto e^{\Delta/k_B T}$.

Problem 1.2 (Density of States in 2D system). Consider free electron in a 2D box with a width of a . The allowed wave-numbers k_x and k_y for x and y directions respectively are expressed as $\frac{\pi n_x}{a}$ and $\frac{\pi n_y}{a}$, where $n_x, n_y = 1, 2, 3, \dots$ are integer numbers. The following figure shows the schematic view of the 2D array of allowed quantum states in k -space.



- (a) Determine the area in k -space occupied by 1 allowed k point.
- (b) Express number of allowed states in the area enclosed by the first quarter circles with the radius of k and $k + dk$, where $k \gg dk$ (Consider the spin degeneracy).
- (c) Express density of allowed states in real space per unit energy.

Solution.

- (a) The area is $\Delta\mathbf{k} = \Delta k_x \cdot \Delta k_y = \left(\frac{\pi}{a}\right)^2$.
- (b) Two electrons can occupy each site

$$dN = 2 \times \frac{dA}{\Delta\mathbf{k}} = \frac{\frac{1}{4}[\pi(k + \Delta k)^2 - \pi k^2]}{(\pi/a)^2} \approx \frac{a^2 k}{\pi} dk.$$

with $|\uparrow\rangle$ and $|\downarrow\rangle$.

- (c) Substituting into the definition of DOS

$$\text{DOS} = \frac{dN}{\Delta A dE} \xrightarrow{E = \hbar^2 k^2 / 2m} \frac{dN/dk}{\Delta A dE/dk} = \frac{m}{\pi \hbar^2}.$$

Problem 1.3 (Density of states from a general formula).

- (a) In a 3D solid and starting from the general formula

$$g(E) = \frac{2V}{(2\pi)^3} \iint \frac{dS_E}{|\nabla_k E|}.$$

Find the expression for the electronic density of states: (i) for free electrons; (ii) for electrons that obey the isotropic dispersion relation: $E = -E_0 - 2\gamma \cos\left(\frac{ka}{2}\right)$, where $E_0, \gamma > 0$.

- (b) After having established the corresponding expression for 2D, find $g(E)$ for (i) and (ii) above.
- (c) Same question for 1D.

Solution. In 3D, 2D, and 1D situations, the DOS are

$$g_{3D}(E) = \frac{2V}{(2\pi)^3} \cdot \frac{4\pi k^2}{\nabla_k E} = \begin{cases} \frac{mV}{\pi^2 \hbar^3} \sqrt{2mE}, \\ \frac{8V}{\pi^2 a^3} \frac{[\arccos(\frac{E+E_0}{-2\gamma})]}{\sqrt{4\gamma^2 - (E+E_0)^2}}. \end{cases}$$
$$g_{2D}(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\nabla_k E} = \begin{cases} \frac{mA}{\pi \hbar^2} = \text{Const}, \\ \frac{4A}{\pi a^2} \frac{\arccos(\frac{E+E_0}{-2\gamma})}{\sqrt{4\gamma^2 - (E+E_0)^2}}. \end{cases}$$
$$g_{1D}(E) = \frac{2L}{2\pi} \cdot \frac{2}{dE/dk} = \begin{cases} \frac{L}{\pi \hbar} \sqrt{\frac{2m}{E}}, \\ \frac{4L}{\pi a} \frac{1}{\sqrt{4\gamma^2 - (E+E_0)^2}}. \end{cases}$$

Homework #2 Phonon

Problem 2.1 (Normal modes of a square lattice). Consider a square lattice of atoms of mass M , in which the nearest neighbors are coupled to each other by springs with constants K . Derive the equation of motion for the mode in which atomic displacements are normal to the crystal plane and find the dispersion of this mode. Analyze the behavior of this mode in the continuum limit.

Solution. Only consider the vibration along the surface. The EOM for the n -th atom is

$$M\ddot{u}_{n,m} = -K \sum [u_{n,m} - u_{n\pm 1, m\pm 1}].$$

Substitute the plane wave solution $u_{n,m}(t) = U e^{i(q_x na + q_y ma - \omega t)}$

$$-M\omega^2 = 2K[\cos(q_x a) + \cos(q_y a) - 2].$$

For $|q_x a|, |q_y a| \ll 1$, $\sin qa = qa$. Then the dispersion of this model becomes linear to $|\mathbf{k}|$

$$\omega(\mathbf{k}) \cong 2\sqrt{\frac{K}{M}} \sqrt{\left(\frac{k_x a}{2}\right)^2 + \left(\frac{k_y a}{2}\right)^2} = a\sqrt{\frac{K}{M}} |\mathbf{k}|.$$

Problem 2.2 (Localized phonons on an impurity). Consider a linear chain of identical atoms of mass m each equidistant by a .

- (a) Limiting the problem to interactions between nearest neighbors characterized by the force constant β , give the dispersion relation of longitudinal phonons and derive the expression of their group velocity. Express the results as a function of “ a ” and of the velocity of sound, v_s . What is maximum frequency of atomic vibration, ω_M , when $v_s = 5000 \text{ m/s}$?
- (b) An atom in the middle of the chain is replaced by the atom of an impurity with a mass m' (where $m' < m$). Write the displacement equations, u_0 , for this impurity atom and the displacement “ u_n ” for an atom located at a distance “ na ”. Assume that the force constant β between nearest neighbors is the same regardless of the atom considered.
- (c) The impurity atom being lighter than the others has an oscillation frequency ω'_m larger than ω_M and it drags the other neighbor atoms by its motion. We try to describe its effect with the expression of a damped wave of form: $u_n = u_0 e^{-\alpha|x|} e^{i(\omega' t - kx)}$. What is the value of the frequency ω'_M , which is compatible with the equations of motion when the wave vector is near

the boundary of the first Brillouin Zone? Verify that when $m = m'$, the solution matches that obtained in (a).

- (d) After having numerically evaluated ω'_M and a when $m' = m/2$, sketch the instantaneous position of atoms as a function of x .

Solution.

- (a) Only consider the vibration along the surface. The EOM for the n -th atom is

$$m\ddot{u}_n = -\beta(2u_n - u_{n+1} - u_{n-1}).$$

Substitute the plane wave solution $u_n(t) = A e^{i(kna - \omega t)}$

$$-m\omega^2 u_n = -\beta(2 - 2\cos(ka))u_n = -4\beta \sin^2(ka/2),$$

we have the dispersion relation $\omega(k) = 2\sqrt{\beta/m} |\sin(ka/2)|$.

Then, the group velocity can be expressed by a and v_s

$$v_g(k) = \frac{d\omega}{dk} = a \sqrt{\frac{\beta}{m}} \cos\left(\frac{ka}{2}\right) \frac{v_s = v_g^{\max}}{v_g^{\max} = a \sqrt{\frac{\beta}{m}}} v_s \cos\left(\frac{ka}{2}\right),$$

as well as the frequency $\omega = \frac{2v_s}{a} |\sin(ka/2)|$. when $k = \pi/a$, the frequency arrives at a maximal value $\omega_M = 2v_s/a$.

- (b) $m' \ddot{u}_0 = \beta(u_1 + u_{-1} - 2u_0)$, $m \ddot{u}_n = \beta(u_{n+1} + u_{n-1} - 2u_n)$.
(c) In (a), near the boundary of the first BZ, i.e. $k = \pi/a$, the frequency is $\omega_1 = 2\sqrt{\beta/m}$. Taking $x = na$

$$u_n = A e^{-\alpha|n|a} e^{i(\omega' t - kna)}.$$

Substitute it into the two EOMs in (b), and let $n = 1$, then derive the two expressions, we have

$$\frac{m'}{m} = \frac{e^{-\alpha a} + 1}{\cosh(\alpha a) + 1} = \frac{2}{e^{\alpha a} + 1}, \quad e^{-\alpha a} + 1 = \frac{2m}{2m - m'}.$$

Substitute it into one of the EOM, we obtain the dispersion

$$\omega_M'^2 = \omega'^2 = 4\beta m / [m'(2m - m')],$$

where $\omega_M'^2 = \omega'^2$ is due to the boundary of the first BZ ($k = \pi/a$). When $m' = m$, the frequency becomes

$$\tilde{\omega}_M'^2 = 4\beta/m, \quad \omega'_M = 2\sqrt{\beta/m} = \omega_1,$$

which matches that obtained in (a).

- (d) When $m' = m/2$, to simplify, taking $t = 0$. Near the first BZ, the displacement of atoms is $u_n = (-1)^n A e^{-\alpha|n|a}$.

Problem 2.3 (Density of states and specific heat of a monoatomic 1D lattice). Consider a row of N identical atoms of mass m and equidistant by a .

- (a) Limiting the problem to interactions between nearest neighbors with a force constant β , find the expression for the dispersion relation of acoustic longitudinal vibrations propagating along this row. What is the maximum frequency ω_M of atomic vibrations, written in terms of v_s and a ? Compare the value obtained with that of ω_D deduced from the Debye model.
(b) Find the corresponding expression for the DOS $g(\omega)$.
(c) Write the expressions for internal energy U and for the heat capacity C_v . Find the limits as these functions approach high temperatures ($\hbar\omega_M \ll k_B T$).
(d) What is the evolution of C_v at low temperature? Compare the result obtained here with that deduced from the Debye model.

Solution.

- (a) The same to Problem 2.2(a): the dispersion

$$\omega(k) = 2\sqrt{\frac{\beta}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|, \quad v_s = a \sqrt{\frac{\beta}{m}}, \quad \omega_M = \frac{2v_s}{a}.$$

The Debye frequency for 1D chain can be obtained from

$$g(\omega) = \frac{1L}{2\pi} \frac{d(2k)}{dk} \left| \frac{dk}{d\omega} \right| = \frac{L}{\pi} \left| \frac{dk}{d\omega} \right| \frac{\omega = v_s k}{L = Na} \frac{Na}{v_s \pi}.$$

Since the DOS satisfies $1N = \int_0^{\omega_D} g(\omega) d\omega = \omega_D Na / v_s \pi$, then we obtained $\omega_D = \frac{\pi v_s}{a}$ with $\omega_M / \omega_D = 2/\pi < 1$.

- (b) Since $dk/d\omega = (d\omega/dk)^{-1} = \frac{a}{2} \sqrt{\omega_M^2 - \omega^2}$, substituting

$$g(\omega) = \frac{L}{\pi} \frac{dk}{d\omega} \frac{L = Na}{\pi \omega_M} \frac{2N}{\sqrt{1 - (\omega/\omega_M)^2}}.$$

- (c) The phonon thermal occupation

$$n(\omega) = (e^{\hbar\omega/k_B T} - 1)^{-1}.$$

Then the internal energy is

$$U(T) = \int_0^{\omega_M} \hbar n(\omega) g(\omega) d\omega = \int_0^{\omega_M} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1} g(\omega) d\omega,$$

and the heat capacity

$$C_V(T) = \frac{dU}{dT} = \int_0^{\omega_M} \frac{\partial n(\omega)}{\partial T} g(\omega) d\omega.$$

At high-temperature limit, $\hbar\omega_M \ll k_B T$. Then

$$U(T) = \int_0^{\omega_M} k_B T g(\omega) d\omega = N k_B T, \quad C_V(T) = N k_B.$$

where the total number of modes $\int_0^{\omega_M} g(\omega) d\omega = N$.

- (d) At low temperature limit, only small k matters, i.e., the main contributions come from small ω . Then, $g(\omega) = g(0) = \frac{2N}{\pi\omega_M}$. The internal energy becomes

$$U \stackrel{x = \frac{\hbar\omega_M}{k_B T}}{=} \frac{2N(k_B T)^2}{\pi \hbar \omega_M} \int_0^\infty \frac{x}{e^x - 1} dx = \frac{N\pi}{3\hbar\omega_M} (k_B T)^2,$$

then the capacity $C_v \propto T$, same as the 1D Debye model.

Homework #3 Crystal lattice

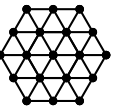
Problem 3.1 (Crystallographic restriction theorem). Proof the crystallographic restriction theorem: Consider a crystal (i.e. with translational symmetry) in three dimensions that is invariant under rotations by an angle $2\pi/n$ around an axis. Then the only allowed possibilities are $n = 1, 2, 3, 4, 6$ with the case $n = 1$ being the trivial case with no rotational symmetry.

Solution. *Proof.* When taking the rotation matrix to an arbitrary vector $\mathbf{l} = (l_1, l_2, l_3)^T$, it should leave the lattice invariant, the trace of the rotation matrix

$$\mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

must be integers, then $\text{Tr}(\mathbf{R}) \in \mathbb{Z}$, $n \in \{1, 2, 3, 4, 6\}$. □

Problem 3.2 (2D triangular lattice). Let us consider a two-dimensional triangular lattice as shown in the following figure.



- (a) Identify the primitive unit cell of the two-dimensional triangular lattice. Find the basis vectors.
(b) Construct the basis vectors of the reciprocal unit cell.

Solution.

- (a) $\mathbf{a}_1 = a(1, 0)$, $\mathbf{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$.
(b) Using the relation $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}$ for $\mathbf{a}_{1,2}$ and $\mathbf{b}_{1,2}$, then $\mathbf{b}_1 = \frac{2\pi}{a}(1, -\frac{1}{\sqrt{3}})$, $\mathbf{b}_2 = \frac{2\pi}{a}(0, \frac{2}{\sqrt{3}})$.

Problem 3.3 (Properties of the reciprocal lattice).

- (a) Show that all reciprocal lattice vectors of the form $\mathbf{G} = h\mathbf{A} + k\mathbf{B} + l\mathbf{C}$ are perpendicular to the planes of the same indices (h, k, l) in real space.

- (b) Show that the distance $d_{h,k,l}$ between two consecutive planes (h, k, l) is inversely proportional to $|\mathbf{G}_{h,k,l}|$.
- (c) Find $d_{h,k,l}$ for a simple cubic lattice and an orthorhombic lattice ($a \neq b \neq c$, $\alpha = \beta = \gamma = \pi/2$).

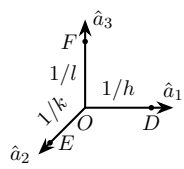
Solution.

- (a) *Proof.* Express $\mathbf{G}$ in terms of the reciprocal lattice vectors
- $$\mathbf{G} = \frac{2\pi}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)} [h(\mathbf{a}_2 \times \mathbf{a}_3) + k(\mathbf{a}_3 \times \mathbf{a}_1) + l(\mathbf{a}_1 \times \mathbf{a}_2)]$$
- with the Miller indices $h : k : l = \frac{1}{a_1} : \frac{1}{a_2} : \frac{1}{a_3}$. In real space, the normal vector of the plane (h, k, l) can be expressed as

$$\hat{\mathbf{n}}_{hkl} \sim \mathbf{r}_1 \times \mathbf{r}_2 = \frac{1}{hkl} [h(\mathbf{a}_2 \times \mathbf{a}_3) + k(\mathbf{a}_3 \times \mathbf{a}_1) + l(\mathbf{a}_1 \times \mathbf{a}_2)],$$

$$\mathbf{r}_1 = \overrightarrow{DE} = \mathbf{a}_2/k - \mathbf{a}_1/h,$$

$$\mathbf{r}_2 = \overrightarrow{EF} = \mathbf{a}_3/l - \mathbf{a}_2/k.$$



Hence, $\hat{\mathbf{n}}_{hkl} \parallel \mathbf{G}$, so $\mathbf{G}$ is perpendicular to (h, k, l) plane. $\square$

- (b) *Proof.* The vectors pointing to the $n+1$ and n -th plane satisfy

$$\mathbf{G} \cdot \mathbf{d}_{n+1} = 2\pi(n+1), \quad \text{and} \quad \mathbf{G} \cdot \mathbf{d}_n = 2\pi n,$$

where $\mathbf{d}_{n+1} - \mathbf{d}_n = d\hat{\mathbf{n}}$. Combine the two relations

$$\mathbf{G} \cdot (\mathbf{r}_{n+1} - \mathbf{r}_n) = \mathbf{G} \cdot \hat{\mathbf{n}}d = 2\pi d.$$

So, we proved $d = 2\pi/(\mathbf{G}_{h,k,l} \cdot \hat{\mathbf{n}}) = 2\pi/|\mathbf{G}_{h,k,l}|$. $\square$

- (c) For a simple cubic lattice

$$\mathbf{a} = (a, 0, 0), \quad \mathbf{b} = (0, b, 0), \quad \mathbf{c} = (0, 0, c).$$

Then, the reciprocal vectors can be expressed as

$$\mathbf{G} = \frac{2\pi h}{a} \hat{\mathbf{x}} + \frac{2\pi k}{b} \hat{\mathbf{y}} + \frac{2\pi l}{c} \hat{\mathbf{z}}.$$

So, we obtain the distance $d = \frac{2\pi}{|\mathbf{G}|} = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right)^{-1/2}$.

Problem 3.4 (Atomic planes and Miller indices: application to lithium). The Bravais lattice of lithium is simple cubic with lattice parameter $a = 3.48 \text{ \AA}$.

- (a) Suppose that the atoms (assumed to be spheres) are placed along the $[111]$, represent the atomic distribution along the following faces (100) , (110) , (111) , and (201) .
- (b) For each such 2D structure, state the direction and basis vector, $\mathbf{a}$ and $\mathbf{b}$ of the elementary lattice as well as the value of the angle, γ .
- (c) Find the atomic concentration and the mass density of lithium ($A \approx 7$).

Solution.

- (a) (b)
- (100) $\mathbf{a} = (0, a, 0)$, $\mathbf{b} = (0, 0, a)$, $\gamma = \langle \mathbf{a}, \mathbf{b} \rangle = 90^\circ$.
- (110) $\mathbf{a} = (a, -a, 0)$, $\mathbf{b} = (0, 0, a)$, $\gamma = \langle \mathbf{a}, \mathbf{b} \rangle = 90^\circ$.
- (111) $\mathbf{a} = (a, 0, -a)$, $\mathbf{b} = (0, a, a)$, $\gamma = \langle \mathbf{a}, \mathbf{b} \rangle = 60^\circ$.
- (201) $\mathbf{a} = (0, a, 0)$, $\mathbf{b} = (a, 0, -2a)$, $\gamma = \langle \mathbf{a}, \mathbf{b} \rangle = 90^\circ$.

- (c) Since the atoms are placed along the $[111]$, then every cubic lattice's corners has an atom, i.e., it is equivalent that there is an atom in every lattice. So, the atomic concentration is

$$n = \frac{1}{V} = \frac{1}{a^3} = 2.3728 \times 10^{28} \text{ atoms/m}^3.$$

The mass density is

$$\rho = nm_{\text{atom}} = nA/N_A = 275.816 \text{ kg/m}^3.$$

Homework #4 X-ray diffraction

Problem 4.1 (Structure factor for a tri-atomic basis). Consider a linear crystal with lattice constant a in which the basis consists of atom of species A at 0 and two atoms of species B at $1/3$ and $2/3$.

- (a) Find the structure factor of such a crystal.
- (b) X-rays of wavelength $\lambda (= a/4)$ irradiate this crystal at an oblique incidence (45°). Using the Ewald construction in the plane of incidence, show graphically, the diffraction directions and indicate the relative weights of the different diffracted intensities.
- (c) State the physical significance of the vector $\mathbf{k}$ from which the extremity is on a plane passing through the origin of the reciprocal lattice.

Solution.

- (a) Substitute $r_i = 0, \frac{a}{3}, \frac{2a}{3}$ into the structure factor

$$S(h) = f_A e^{i\mathbf{G} \cdot 0} + f_B e^{i\mathbf{G} \cdot \frac{a}{3}} + f_B e^{i\mathbf{G} \cdot \frac{2a}{3}}$$

$$\frac{\text{Euler's formula}}{G=2\pi h/a} \begin{cases} f_A + 2f_B, & h = 3n, \\ f_A - f_B, & \text{otherwise.} \end{cases}$$

- (b) The incident wave number and the wavevector at 45°

$$k_i = 2\pi/\lambda = 8\pi/a, \quad \mathbf{k}_i = \left(k/\sqrt{2}, -k/\sqrt{2} \right).$$

Apply the Laue condition $\mathbf{k} - \mathbf{k}_i = \mathbf{G}$,

then project to the cristal line

$$|\mathbf{k}_i| \cos \theta - |\mathbf{k}_i| \cos 45^\circ = 2\pi h/a$$

one can obtain $\cos \theta = (h + 2\sqrt{2})/4 \in$

$[-1, 1]$, $h = -6, \dots, 0, 1$.

The intensity of each diffraction order is proportional to $|S(h)|^2$, i.e.,

$$I_h \propto |S(h)|^2 = \begin{cases} (f_A + 2f_B)^2, & h = 3m, \\ (f_A - f_B)^2, & \text{otherwise.} \end{cases}$$

- (c) The Bragg condition is satisfied for diffraction.

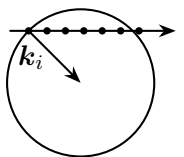
Problem 4.2. The main experimental methods of X-ray diffraction are

- (a) the Laue method (b) the crystal rotation method
- (c) the powder method (or the Debye-Scherrer method)

Describe the corresponding experimental set-up and with the help of the Ewald construction in reciprocal space, deduce the observed diffraction patterns in each case.

Solution.

- (a) Laue Method
- Setup: Stationary single crystal, polychromatic X-rays.
 - Ewald & Pattern: A continuous range of Ewald sphere radii (due to varying λ) means many reciprocal lattice points lie on some sphere. This produces a pattern of sharp spots, each from a different (hkl) plane and wavelength. The spot arrangement directly reveals the crystal's symmetry.
- (b) Crystal Rotation Method
- Setup: Rotating single crystal, monochromatic X-rays.



- ii. Ewald & Pattern: The fixed Ewald sphere is intersected by rotating reciprocal lattice points. This generates discrete spots or arcs on the detector. Measuring the rotation angles and intensities of these spots allows for the determination of the unit cell and atomic structure.

(c) Powder Method

- i. Setup: Powdered sample (random crystallites), monochromatic X-rays.
- ii. Ewald & Pattern: Each set of (hkl) planes forms a spherical shell in reciprocal space. The Ewald sphere's intersection with these shells creates cones of diffracted beams (Debye cones), which appear as characteristic rings on the detector. The ring positions (2θ angles) provide d -spacings for phase identification and lattice parameter calculation.

Homework #5 Nearly free electron model

Problem 5.1 (Compare Kronig-Penney model with the nearly free electron model). The Kronig-Penney model allows one to consider both the nearly-free-electron and tight-binding limits. Consider the (repulsive) Dirac-Kronig-Penney model. It gives the following implicit equation for the energy bands

$$\cos(ka) = \cos(qa) + u \sin(qa)/qa,$$

where $q = \sqrt{2mE}/\hbar$ and $u = \frac{mU_0a}{\hbar^2} > 0$. Assuming that $u \ll q$, find the solution of above equation near the Bragg points $k \approx \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \dots$. Compare your result with that of the nearly-free electron model. (Reminder: the $\approx$ sign means that k is close yet not equal to the wavenumber of the Bragg point.)

Solution. Near the Bragg points, assume

$$ka = n\pi + \kappa a, \quad \text{and} \quad qa = q_n a + \delta.$$

with $|\kappa|, |\delta| \ll 1$, $q_n = n\pi/a$. Keep the 2nd order of κ and δ , the implicit equation can be written as

$$(-1)^n \left[1 - \frac{1}{2}(\kappa a)^2 \right] = (-1)^n \left(1 - \frac{1}{2}\delta^2 \right) + u \frac{(-1)^n \delta}{n\pi}.$$

Then, we can have the solution of δ

$$\delta = u/n\pi \pm \sqrt{(u/n\pi)^2 + (\kappa a)^2}.$$

Since $q = \sqrt{2mE}/\hbar$, then we have

$$E = \hbar^2(q_n + \delta/a)^2/2m \approx E_n^0 + \hbar^2 q_n \delta / ma,$$

where $E_n^0 = \frac{\hbar^2 q_n^2}{2m}$. Substitute $q_n = n\pi/a$, and the solution of δ into the expression of E

$$E = E_n^0 + \frac{n\pi\hbar^2\delta}{m} = E_n^0 + \frac{n\pi\hbar^2}{ma^2} \left[\frac{u}{n\pi} \pm \sqrt{\left(\frac{u}{n\pi}\right)^2 + (\kappa a)^2} \right].$$

Substitute $u = mU_0a/\hbar^2$ into the expression above

$$E = E_n^0 + \frac{U_0}{a} \pm \sqrt{\left(\frac{U_0}{a}\right)^2 + \left(\frac{n\pi\hbar^2\kappa}{ma}\right)^2}$$

with the band gap

$$\Delta E_n = E_+ - E_- = 2\sqrt{\left(\frac{U_0}{a}\right)^2 + \left(\frac{n\pi\hbar^2\kappa}{ma}\right)^2}.$$

When $\kappa a \rightarrow 0$, the energy band and the band gap become

$$E = E_n^0 + U_0/a \pm U_0/a, \quad \text{and} \quad \Delta E_n = 2U_0/a.$$

The results perfectly match that of the nearly-free electron model.

Problem 5.2 (Nearly free electron model). Consider 2D electrons subject to a weak periodic potential

$$U(x, y) = U_0[\cos(2\pi x/a) + \cos(2\pi y/a)].$$

- (a) Find the energy bands near the *edges* of the first Brillouin zone. (Hint: the edges of the Brillouin zone are the Bragg planes where the eigenstates are doubly degenerate.) Plot the bands and isoenergetic surfaces.
- (b) Repeat the analysis of part (a) for regions near the *corners* of the first Brillouin zone, where there are four degenerate eigenstates.

Solution.

- (a) The reciprocal vectors are $\mathbf{G}_1 = \left(\frac{2\pi}{a}, 0\right)$, $\mathbf{G}_2 = \left(0, \frac{2\pi}{a}\right)$. Apply Fourier transformation to the potential

$$U(x, y) = \frac{U_0}{2} \left[e^{\pm i\mathbf{G}_1 \cdot \mathbf{r}} + e^{\pm i\mathbf{G}_2 \cdot \mathbf{r}} \right] = \sum_{\mathbf{G}} U_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}}.$$

So, the Fourier coefficient is $U_{\pm\mathbf{G}_{1,2}} = U_0/2$. Near the zone edge (e.g., $k_x = \pi/a$), consider a wavevector

$$\mathbf{k} = (\pi/a + q_x, k_y)$$

with $|q_x| \ll \pi/a$. The two nearly degenerate plane waves are $|\mathbf{k}\rangle$ and $|\mathbf{k} - \mathbf{G}_1\rangle$. Their free-electron energies are

$$\begin{aligned} E^{(0)}(\mathbf{k}, \mathbf{k} - \mathbf{G}_1) &= \frac{\hbar^2}{2m} \left[\left(\pm \frac{\pi}{a} + q_x \right)^2 + k_y^2 \right] \\ &\approx E_0 \pm \alpha q_x = \frac{\hbar^2}{2m} \left[\left(\frac{\pi}{a} \right)^2 + k_y^2 \right] \pm \frac{\hbar^2 \pi}{ma} q_x. \end{aligned}$$

The perturbation couples these states with matrix element $\langle \mathbf{k} | U | \mathbf{k} - \mathbf{G}_1 \rangle = U_0/2$. In the subspace $\{|\mathbf{k}\rangle, |\mathbf{k} - \mathbf{G}_1\rangle\}$, the Hamiltonian is

$$H = \begin{pmatrix} \langle \mathbf{k} | & | \mathbf{k} - \mathbf{G}_1 \rangle \\ \langle \mathbf{k} - \mathbf{G}_1 | & \end{pmatrix} \begin{pmatrix} E_0 + \alpha q_x & U_0/2 \\ U_0/2 & E_0 - \alpha q_x \end{pmatrix}.$$

From the eigenfunction $\det(\mathcal{H} - E\mathbb{1}) = 0$, one can obtain the eigenvalues

$$E_{\pm} = E_0 + \frac{\hbar^2 q_x^2}{2m} \pm \sqrt{\left(\frac{\hbar^2 \pi}{ma} q_x \right)^2 + \left(\frac{U_0}{2} \right)^2}.$$

- (b) The corners (*M*-point) of the first Brillouin zone are

$$\mathbf{k}_1 = \mathbf{k}_0 = (+\pi/a, +\pi/a),$$

$$\mathbf{k}_2 = \mathbf{k}_0 - \mathbf{G}_1 = (-\pi/a, +\pi/a),$$

$$\mathbf{k}_3 = \mathbf{k}_0 - \mathbf{G}_2 = (+\pi/a, -\pi/a),$$

$$\mathbf{k}_4 = \mathbf{k}_0 - \mathbf{G}_1 - \mathbf{G}_2 = (-\pi/a, -\pi/a),$$

with the same free-electron energies

$$E_0 = \frac{\hbar^2}{2m} (k_x^2 + k_y^2) = \frac{\hbar^2 \pi^2}{ma^2}.$$

In the basis $\{|\mathbf{k}_1\rangle, |\mathbf{k}_2\rangle, |\mathbf{k}_3\rangle, |\mathbf{k}_4\rangle\}$, states are coupled if they differ by $\pm\mathbf{G}_1$ or $\pm\mathbf{G}_2$. The non-zero off-diagonal matrix elements are

$$\langle \mathbf{k}_1 | U | \mathbf{k}_2 \rangle = \langle \mathbf{k}_1 | U | \mathbf{k}_3 \rangle = \langle \mathbf{k}_2 | U | \mathbf{k}_4 \rangle = \langle \mathbf{k}_3 | U | \mathbf{k}_4 \rangle = \frac{U_0}{2}.$$

The Hamiltonian is

$$\mathcal{H} = E_0 \mathbb{1} + \frac{U_0}{2} \begin{pmatrix} \langle \mathbf{k}_1 | & \langle \mathbf{k}_2 | & \langle \mathbf{k}_3 | & \langle \mathbf{k}_4 | \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \end{pmatrix}.$$

From the eigenfunction $\det(\mathcal{H} - E\mathbb{1}) = 0$, one can obtain the

eigenvalues $E = \{E_0 + U_0, E_0 - U_0, E_0, E_0\}$.

Homework #6 Tight binding model

Problem 6.1 (Tight bindings in the BCC and FCC lattices). We consider identical atoms located at points of a lattice that are body-centered cubic (**BCC: i**) and face-centered cubic (**FCC: ii**).

- (a) For each case, determine the dispersion relation for s -electrons, having the general form

$$E = -\alpha - \gamma \sum_s e^{-i\mathbf{k} \cdot \mathbf{R}_s},$$

where α and γ are the positive energies that can be calculated and $\mathbf{R}_s$ represents the vectors that connect an atom located at the origin to each of its nearest neighbors.

- (b) Find the expressions for E at the points located at the intersection of the first Brillouin zone with the directions $[100]$, $[110]$, and $[111]$ and thus the value $E(\Gamma)$ to the center of this zone.
- (c) Deduce the total width of the corresponding band, ΔE , and find the expression for the effective mass m^* in the vicinity of Γ as well as the shape of constant energy surfaces around this point.

Solution (i = BCC, ii = FCC). Assume the lattice constant a .

- (a_i) Each atom has 8 NNs, i.e., $\{\mathbf{R}_s\} = \{\frac{a}{2}(\pm 1, \pm 1, \pm 1)\}$. The dispersion relation is

$$E(\mathbf{k}) = -\alpha - 8\gamma \cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{ak_y}{2}\right) \cos\left(\frac{ak_z}{2}\right).$$

- (b_i) From the relation can obtain the vectors $\mathbf{k}$.

$$|\mathbf{k}|^2 = |\mathbf{k} - \mathbf{G}|^2 = |\mathbf{k}|^2 - 2\mathbf{k} \cdot \mathbf{G} + |\mathbf{G}|^2, \quad 2\mathbf{k} \cdot \mathbf{G} = |\mathbf{G}|^2.$$

- i. Γ point: $\mathbf{k} = (0, 0, 0)$, $E(\Gamma) = -\alpha - 8\gamma$.
- ii. $[100]$: $\mathbf{k} = \frac{2\pi}{a}(\pm 1, 0, 0)$, $E(X) = -\alpha \pm 8\gamma$.
- iii. $[110]$: $\mathbf{k} = \frac{2\pi}{a}(\frac{1}{2}, \frac{1}{2}, 0)$, $E(N) = -\alpha$.
- iv. $[111]$: $\mathbf{k} = \frac{\pi}{a}(1, 1, 1)$, $E(P) = -\alpha$.

- (c_i) The cosine terms in the general form differs between $[-8\gamma, +8\gamma]$. So, the energy gap is

$$\Delta E_{\text{BCC}} = (-\alpha + 8\gamma) - (-\alpha - 8\gamma) = 16\gamma.$$

The dispersion relation can be expanded around the Γ point

$$E(\mathbf{k}) = -\alpha - 8\gamma \left[1 - \sum_i \frac{1}{2} \left(\frac{ak_i}{2} \right)^2 \right] = -\alpha - 8\gamma + \gamma a^2 k^2.$$

Then, the effective mass $m^* = \frac{\hbar^2}{\partial^2 E / \partial k^2} = \frac{\hbar^2}{2\gamma a^2}$.

- (a_{ii}) Each atom has 12 NNs, i.e.,

$$\{\mathbf{R}_s\} = \left\{ \frac{a}{2}(\pm 1, \pm 1, 0), \frac{a}{2}(\pm 1, 0, \pm 1), \frac{a}{2}(0, \pm 1, \pm 1) \right\}.$$

The dispersion relation is

$$E(\mathbf{k}) = -\alpha - 4\gamma \left[\cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{ak_y}{2}\right) + \text{Cyclic} \right].$$

- (b_{ii}) i. Γ point: $\mathbf{k} = (0, 0, 0)$, $E(\Gamma) = -\alpha - 12\gamma$.
- ii. $[100]$: $\mathbf{k} = \frac{2\pi}{a}(1, 0, 0)$, $E(X) = -\alpha + 4\gamma$.
- iii. $[110]$: $\mathbf{k} = \frac{2\pi}{a}(\frac{3}{4}, \frac{3}{4}, 0)$, $E(N) = -\alpha + \gamma(4\sqrt{2} - 2)$.
- iv. $[111]$: $\mathbf{k} = \frac{2\pi}{a}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $E(P) = -\alpha$.

- (c_{ii}) The cos terms in the general form differs between $[-12\gamma, +4\gamma]$.

So, the energy gap is

$$\Delta E_{\text{FCC}} = (-\alpha + 4\gamma) - (-\alpha - 12\gamma) = 16\gamma.$$

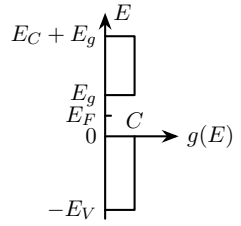
The dispersion relation can be expanded around the Γ point

$$E(\mathbf{k}) = -\alpha - 4\gamma \sum_{\text{cyclic}(x,y,z)} \left[1 - \frac{1}{2} \left(\frac{ak_i}{2} \right)^2 - \frac{1}{2} \left(\frac{ak_j}{2} \right)^2 \right] = -\alpha - 12\gamma + \gamma a^2 k^2.$$

Then, the effective mass $m^* = \frac{\hbar^2}{\partial^2 E / \partial k^2} = \frac{\hbar^2}{2\gamma a^2}$.

Homework #7 Semiconductor

Problem 7.1 (Elementary study of an intrinsic semiconductor). Consider a semiconductor characterized by a constant density of states ($g(E) = C$) in both the valence and conduction band. These bands, separated by E_g , have a respective width of E_V and E_C (see the Figure).



- (a) Knowing that the Fermi level E_F is in the bandgap ($5k_B T < E_F < E_g - 5k_B T$), leading to the conventional simplification for $f(E)$ (The Boltzmann approximation can be applied), find the expression for the number of electrons n in the conduction band and the number of holes h in the valence band respectively as a function of the data and absolute temperature.
- (b) Deduce the expression for the volumetric density of intrinsic carriers n_i at ambient temperature ($k_B T = 25 \text{ meV}$) as well as the position of the Fermi level E_{F_i} .
Numerical applications: $E_g = 0.7 \text{ eV}$, $E_C = E_V = 5 \text{ eV}$, and $C = 2 \times 10^{21} \text{ electrons/cm}^3/\text{eV}$.
- (c) At the same temperature, what is the intrinsic electronic conductivity of this material? Take $\mu_h = \mu_e = 1000 \text{ cm}^2/\text{Vs}$.
- (d) The material is doped with N_0 boron impurities. Find the expressions and the new values of n_d , h_d , σ_d , and E_{F_d} with $N_0 = 10^{12} \text{ impurities per cm}^3$ and next with $N_0 = 10^{14} \text{ impurities per cm}^3$.

Solution.

- (a) The number of electrons in the conduction band

$$n = \int_{E_g}^{E_g + E_C} g(E) f(E) dE \stackrel{\text{B. approx.}}{\approx} C k_B T e^{-\frac{E_g - E_F}{k_B T}}.$$

The number of holes in the valence band

$$h = \int_{-E_V}^0 g(E) [1 - f(E)] dE \stackrel{\text{B. approx.}}{\approx} C k_B T e^{-\frac{E_F}{k_B T}}.$$

where $f(E) \approx e^{-\frac{E - E_F}{k_B T}}$, $[1 - f(E)] \approx e^{\frac{E - E_F}{k_B T}}$.

- (b) For the intrinsic carriers, the number of electrons and holes are the same

$$n = C k_B T e^{-\frac{E_g - E_F}{k_B T}} = h = C k_B T e^{-\frac{E_F}{k_B T}}, \quad E_F = E_g/2.$$

Substitute it back into the expression of n (or h), we get the intrinsic carrier concentration

$$n_i = C k_B T e^{-\frac{E_g}{2k_B T}} = 4.158 \times 10^{19} \text{ m}^{-3}, \quad E_F = 0.35 \text{ eV}.$$

- (c) The conductivity $\sigma_i = e(n_i \mu_e + n_i \mu_h) = 1.332 \text{ S/m}$.
- (d) Boron is an acceptor, so doping concentration $N_A = N_0$. Now, the number of electrons and holes

$$n = n_i, \quad h = n_i \rightarrow n_d = n, \quad h_d = n + N_0,$$

and they satisfy the identity: mass-action law

$$n_i^2 = n_d h_d = n(n + N_0).$$

We can solve that

$$n_d = \frac{\sqrt{N_0^2 + 4n_i^2} - N_0}{2}, \quad h_d = N_0 + \frac{\sqrt{N_0^2 + 4n_i^2} - N_0}{2}.$$

The conductivity becomes $\sigma_d = e(n_d\mu_e + h_d\mu_h)$. From $h = Ck_B T e^{-\frac{E_F}{k_B T}}$, we can obtain the Fermi level

$$E_{F_d} = k_B T \ln(Ck_B T/h_d).$$

Homework #8 Electrons in a magnetic field

Problem 8.1 (Onsager formula). Suppose that the electron spectrum for a tetragonal lattice can be described by the following equation

$$E_k = \frac{\hbar^2 k_x^2}{2m_{xx}} + \frac{\hbar^2 k_y^2}{2m_{xx}} + \frac{\hbar^2 k_z^2}{2m_{zz}},$$

with $m_{xx} \neq m_{zz}$. Using the Onsager formula for the cyclotron mass, $m_c = \frac{\hbar^2}{2\pi} \frac{\partial A}{\partial E_k}$ with A being the area of an isoenergetic contour perpendicular to the magnetic field. Find m_c for the case when the magnetic field is

- (a) Along the z -axis. (b) In the $x - y$ plane.

Solution.

- (a) For $\mathbf{B} \parallel \hat{z}$, the motion of electrons will perpendicular to $\hat{z}$, i.e., parallel to $x - y$ plane. The equation of motion becomes

$$E_k = \frac{\hbar^2}{2m_{xx}}(k_x^2 + k_y^2), \quad \frac{k_x^2}{2m_{xx}E_k/\hbar^2} + \frac{k_y^2}{2m_{xx}E_k/\hbar^2} = 1.$$

Substituting to the definition of cyclotron mass

$$A = \pi(k_{x0}^2 + k_{y0}^2) = \frac{2\pi m_{xx}E_k}{\hbar^2}, \quad m_c = \frac{\hbar^2}{2\pi} \frac{\partial A}{\partial E_k} = m_{xx}.$$

- (b) Simply taking $\mathbf{B} \parallel \hat{x}$, then orbits lie in the $k_y - k_z$ plane with an ellipse area. The equation of motion becomes

$$E_k = \frac{\hbar^2 k_y^2}{2m_{xx}} + \frac{\hbar^2 k_z^2}{2m_{zz}}, \quad \frac{k_y^2}{2m_{xx}E_k/\hbar^2} + \frac{k_z^2}{2m_{zz}E_k/\hbar^2} = 1.$$

Similarly to (a)

$$A = \pi k_{y0} k_{z0} = 2\pi E_k \sqrt{m_{xx}m_{zz}}/\hbar^2, \quad m_c = \sqrt{m_{xx}m_{zz}}.$$

Problem 8.2 (Integer Quantum Hall Effect). The Hall resistance R_{xy} of a two-dimensional electron gas is measured as a function of magnetic field B at a very low temperature. A series of plateaus are observed at values $R_{xy} = \frac{h}{(\nu e^2)}$, where ν is an integer. Simultaneously, the longitudinal resistance R_{xx} shows peaks and drops to zero.

- (a) Explain physically why R_{xx} drops to zero whenever R_{xy} is on a plateau.
(b) A sample has an electron density of $n_s = 4.0 \times 10^{11} \text{ cm}^{-2}$.
i. Calculate the filling factor ν at $B = 2.0 \text{ T}$.
ii. What is the expected value of R_{xy} on the plateau nearest to this field?
iii. How many Landau levels are completely filled at this plateau?
(c) Why is the IQHE used as a resistance standard? Which two fundamental constants are involved?

Solution.

- (a) When the Hall resistance R_{xy} is on a plateau, the Fermi energy lies in a mobility gap between Landau levels. All extended states in the bulk are absent, making the bulk insulating. Current flows only along dissipationless chiral edge states. At the edges, backscattering is prevented by spatial separation: even if impurities or imperfections exist along an edge,

they may cause small forward deviations in an electron's path but cannot reverse its direction, so no resistance is produced. With no extended states in the bulk to allow dissipation and no backscattering at the edges, the longitudinal voltage drop vanishes, giving $R_{xx} = 0$.

- (b) i. From the degeneracy of Landau level $n_s = \nu \frac{eB}{h}$, we can obtain the filling factor $\nu = \frac{n_s h}{eB} = 8.271$.
ii. Substituting the values into the expression of the Hall resistance $R_{xy} = R_K/\tilde{\nu} = h/\tilde{\nu}e^2 = 3226.60 \Omega$, where $\tilde{\nu}$ is the nearest integer of ν .
iii. The filling factor $\tilde{\nu} = 8$ directly corresponds to the number of completely filled Landau levels.
(c) IQHE provides a quantized Hall resistance $R_H = h/e^2$, which is precisely quantized and only based on two fundamental constants: The Planck constant h and the elementary charge e , so it is used as a resistance standard.

Problem 8.3 (Lifshitz-Kosevich formula). The Lifshitz-Kosevich formula describes the amplitude of the Shubnikov-de Haas (SdH) oscillations. The amplitude ΔR is proportional to the factor: $R_T = \lambda(T)/\text{sh } \lambda(T)$, where $\lambda(T) = 2\pi^2 k_B T m^*/\hbar e B$.

- (a) A researcher measures the SdH oscillation amplitude ΔR at a fixed magnetic field B as a function of temperature T . How can they use this data to determine the effective mass m^* of the charge carriers?
(b) Another researcher fixes the temperature T and measures the SdH amplitude ΔR as a function of $1/B$. They find that the amplitude decreases exponentially with $1/B$. What physical parameter does this decay reveal, and what does a rapid decay imply about the quality of the sample?

Solution.

- (a) Take $\alpha = 2\pi^2 k_B m^*/\hbar e B$, then $\lambda(T) = \alpha T$. Since at $T \gg 1$, the amplitude ΔR satisfies $\Delta R \propto R_T = \alpha T/\text{sh}(\alpha T) \rightarrow \alpha T e^{-\alpha T}/2$, $\Delta R = \tilde{C} T e^{-\alpha T}$. Take the logarithm to the two sides

$$\ln(\Delta R/T) = -\alpha T + C,$$

where $C = \ln \tilde{C}$. So, one can take plot the linear relation. By fitting the slope $k = -\alpha$ that fits the figure, one can obtain the effective mass

$$m^* = \alpha \hbar e B / 2\pi^2 k_B = -k \hbar e B / 2\pi^2 k_B.$$

- (b) Similarly to part (a), we take $T \gg 1$. Since ΔR is proportional to $R_T R_D R_S$, and if we only consider the magnetic variable B , then $\Delta R \propto R_T R_D$. To be more specific

$$\Delta R \propto \frac{2\pi^2 k_B T m^*}{\hbar e} \cdot \frac{1}{B} \cdot e^{-\frac{2\pi^2 k_B T m^*}{\hbar e} \cdot \frac{1}{B}} \cdot e^{-\frac{\pi m^*}{e \tau_q} \cdot \frac{1}{B}} = \tilde{C} \cdot e^{-(\alpha + \beta/\tau_q) \cdot \frac{1}{B}} / B,$$

where the coefficients α and β can be easily known since we have already obtained the effective mass. Taking the logarithm to both sides, we can obtain

$$\ln(\Delta R) = C - \ln B - (\alpha + \beta/\tau_q)/B,$$

Since the effective mass m^* has been determined from (a), so the coefficients α and β are known, and the slope directly reveals the quantum scattering rate $1/\tau_q$.

A rapid exponential decay corresponds to a short τ_q , meaning a short mean free path and therefore significant disorder: an indicator of poor sample quality.

Homework #9 Band structure of graphene

Problem 9.1. In the calculation of the monolayer graphene band structure using tight-binding model, if we keep the contribution of the next-nearest hopping t' .

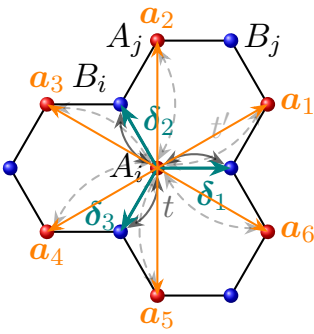
- Write down the dispersion relation.
- Expand the energy spectrum around the Dirac point up to second order. Search the literature and check what's trigonal warping effect is in a monolayer.

Solution.

- The Hamiltonian in second quantization form for the graphene with the next-nearest (NN) hopping is

$$\mathcal{H} = -t \sum_{\langle i,j \rangle} (\hat{a}_i \hat{b}_j^\dagger + \text{H.c.}) - t' \sum_{\langle\langle i,j \rangle\rangle} (\hat{a}_i \hat{a}_j^\dagger + \hat{b}_i \hat{b}_j^\dagger + \text{H.c.}),$$

where i and j label the nearest neighbor, i.e., the bond A_i, B_i . The fermion operators a_i and $a_i^\dagger$ annihilate/create an electron at site A , also for b_i and $b_i^\dagger$ at site B . The vectors pointing from A_i to the 3 nearest B -type sites and 6 next-nearest A -type sites are



$$\begin{aligned} \delta_1 &= a(1, 0), \\ \delta_2 &= a(-1/2, \sqrt{3}/2), \\ \delta_3 &= a(-1/2, -\sqrt{3}/2); \\ a_1 &= a(3/2, \sqrt{3}/2), \\ a_2 &= a(0, \sqrt{3}), \\ a_3 &= a(-3/2, \sqrt{3}/2), \\ a_4 &= a(-3/2, -\sqrt{3}/2), \\ a_5 &= a(0, -\sqrt{3}), \\ a_6 &= a(3/2, -\sqrt{3}/2). \end{aligned}$$

Assume there are $2N$ atoms in total, then there are N A -type or B -type sites in total. Applying the Fourier transformation to the fermion operators

$$\begin{aligned} \hat{a}_i &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} \hat{a}_{\mathbf{k}}, & \hat{b}_{i+\delta} &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, j} e^{i\mathbf{k} \cdot (\mathbf{r}_i + \delta_j)} \hat{b}_{\mathbf{k}}, \\ \hat{a}_i^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} \hat{a}_{\mathbf{k}}^\dagger, & \hat{b}_{i+\delta}^\dagger &= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, j} e^{-i\mathbf{k} \cdot (\mathbf{r}_i + \delta_j)} \hat{b}_{\mathbf{k}}^\dagger. \end{aligned}$$

Consider the first term of the Hamiltonian, it can be expressed as

$$\begin{aligned} \mathcal{H}_t &= -t \sum_{i \in A} \sum_{j=1}^3 \left(\frac{1}{N} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} \hat{a}_{\mathbf{k}}^\dagger \sum_{\mathbf{k}'} e^{i\mathbf{k}' \cdot (\mathbf{r}_i + \delta_j)} \hat{b}_{\mathbf{k}'} \right. \\ &\quad \left. + \text{h.c.} \right) = -t \sum_{\mathbf{k}} \sum_{j=1}^3 e^{i\mathbf{k} \cdot \delta_j} \hat{a}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{k}} + \text{h.c.} \end{aligned}$$

where $\sum_{i \in A} e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}_i} = N \delta_{\mathbf{k}', \mathbf{k}}$ is applied, and multiple sum will not occur here since we only sum over every A -type site.

Denote $\Delta(\mathbf{k}) = -t \sum_{j=1}^3 e^{i\mathbf{k} \cdot \delta_j}$, this part becomes

$$\mathcal{H}_t = \sum_{\mathbf{k}} \Delta(\mathbf{k}) \hat{a}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{k}} + \sum_{\mathbf{k}} \Delta^*(\mathbf{k}) \hat{b}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}.$$

For the second part $\mathcal{H}_{t'}$: Denote $\epsilon(\mathbf{k}) = -t' \sum_{m=1}^6 e^{i\mathbf{k} \cdot \mathbf{a}_m}$, one can obtain (The Hermit conjugate is applied, but the factor $\frac{1}{2}$ is also applied to prevent a twice sum between the same NN

bond.)

$$\mathcal{H}_{t'} = \sum_{\mathbf{k}} \epsilon(\mathbf{k}) \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \sum_{\mathbf{k}} \epsilon(\mathbf{k}) \hat{b}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{k}}.$$

Here the $\epsilon(\mathbf{k})$ should not taken Hermit conjugate, since the hopping $A_i - A_j$ and $B_i - B_j$ are the same physical process. Hence, the total Hamiltonian is

$$\mathcal{H} = \sum_{\mathbf{k}} \underbrace{\begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{b}_{\mathbf{k}} \end{pmatrix}}_{\Psi^\dagger} \underbrace{\begin{pmatrix} \epsilon(\mathbf{k}) & \Delta(\mathbf{k}) \\ \Delta^*(\mathbf{k}) & \epsilon(\mathbf{k}) \end{pmatrix}}_{\mathbf{H}_{2 \times 2}} \underbrace{\begin{pmatrix} \hat{a}_{\mathbf{k}} \\ \hat{b}_{\mathbf{k}} \end{pmatrix}}_{\Psi}.$$

By taking $\det(\mathbf{H}_{2 \times 2} - E\mathbf{1}) = 0$, one can obtain the dispersion relation

$$(\epsilon(\mathbf{k}) - E)^2 = \Delta(\mathbf{k}) \Delta^*(\mathbf{k}).$$

To obtain the exact expression, expanding $\Delta(\mathbf{k})$ and $\epsilon(\mathbf{k})$

$$\Delta(\mathbf{k}) \Delta^*(\mathbf{k}) = t^3 (3 + f(\mathbf{k})), \quad \epsilon(\mathbf{k}) = -t' f(\mathbf{k})$$

with $f(\mathbf{k}) = 2 \cos(\sqrt{3} k_y a) + 4 \cos(3 k_x a/2) \cos(\sqrt{3} k_y a/2)$. Substituting them back to the dispersion relation, we can obtain $E_{\pm} = \pm t \sqrt{3 + f(\mathbf{k})} - t' f(\mathbf{k})$.

- Take the Dirac point $\mathbf{K} = \frac{2\pi}{3\sqrt{3}a}(\sqrt{3}, 1)$, then take $|\mathbf{q}| \ll 1$

$$f(\mathbf{K} + \mathbf{q}) = -3 + \frac{9}{4} |\mathbf{q}|^2 a^2,$$

$$\text{Now } E_{\pm} = \pm \frac{3t}{2} |\mathbf{q}| a + 3t' - \frac{9}{4} t' |\mathbf{q}|^2 a^2 + \mathcal{O}(|\mathbf{q}|^3).$$

Problem 9.2. Graphene has the record room-temperature electron mobility, mainly due to two reasons. The first is the high Debye temperature (~ 2100 K) that suppresses phonon scattering arising from the in-plane covalent sp^2 bonding between carbon atoms. The second one is related to its unique linear Dirac band, resulting in massless Dirac fermions ($m^* = 0$) near the Dirac point. However, if we apply the definition of the effective mass $m^* = \hbar^2 (d^2 E(k)/dk^2)^{-1}$ to the band dispersion of graphene ($E_{\pm}(k) = \pm \hbar v_F k$), we will get diverging m^* and non-differentiable Dirac point. Please explain this contradiction.

Solution. The contradiction arises from applying different definitions of “mass” to graphene’s unique band structure. The description of “massless Dirac fermions” comes from the analogy to relativistic particles.

From a massless special relativity particle like a photon to the electrons in the graphene, their dispersion relations have the comparison $E = pc \implies E = \pm \hbar v_F k = p v_F$ with v_F playing the role of c . Hence, electrons near the Dirac point behave as if they have zero rest mass.

However, the conventional effective mass $m^* = \hbar^2 (d^2 E/dk^2)^{-1}$ is defined under the parabolic-band approximation $E \propto k^2$. For graphene’s linear dispersion, $d^2 E/dk^2 = 0$ away from $k = 0$, leading to $m^* \rightarrow \infty$ mathematically. At $k = 0$ the dispersion is non-differentiable, so the formula breaks down.

Physically, this infinite effective mass reflects that the electron’s speed is constant at v_F : An external force parallel to its momentum cannot change its speed, much like a photon’s speed is fixed at c . Thus, the “massless” label refers to the relativistic analogy, while the divergent m^* indicates the inapplicability of the parabolic effective mass concept to a linear band.

II. TIGHT-BINDING APPROACH TO STRAINED KAGOME LATTICE

In this section, we investigate the band structures of strained kagome lattice using the tight-binding approximation. Since the band structures of kagome lattice are primarily characterized by the coexistence of Dirac cones and a flat band, here we mainly focus on the shift of the Dirac cones and the reshaping of the flat band in response to applied uniaxial strain.

A. Tight-binding model for kagome lattice

Fig. 1(a) is an illustration of the nearest-neighbor tight-binding model for strained kagome lattice. The kagome lattice is a triangular Bravais lattice with a unit cell composed of three inequivalent sites, which are labeled as A , B and C , respectively. The three sites in the unit cell form a regular triangle, as indicated by the shaded region in Fig. 1(a). The repeated Brillouin zone of kagome lattice is shown in Fig. 1(b), where the first Brillouin zone is a regular hexagon with two independent corners K and K' .

The tight-binding Hamiltonian of kagome lattice is given as [11, 34]

$$H = - \sum_{\mathbf{r}} [t_1(a_{\mathbf{r}}^\dagger b_{\mathbf{r}} + a_{\mathbf{r}}^\dagger b_{\mathbf{r}+\delta_1}) + t_2(a_{\mathbf{r}}^\dagger c_{\mathbf{r}} + c_{\mathbf{r}}^\dagger a_{\mathbf{r}+\delta_2}) + t_3(b_{\mathbf{r}}^\dagger c_{\mathbf{r}} + c_{\mathbf{r}}^\dagger b_{\mathbf{r}+\delta_3})] + \text{h.c.}, \quad (1)$$

with the summation of $\mathbf{r}$ running over all unit cells. Here $a_{\mathbf{r}}^\dagger$, $b_{\mathbf{r}}^\dagger$ and $c_{\mathbf{r}}^\dagger$ ($a_{\mathbf{r}}$, $b_{\mathbf{r}}$, and $c_{\mathbf{r}}$) are respectively the creation (annihilation) operators of electrons at sites A , B and C in the unit cell located at $\mathbf{r}$, and t_i ($i = 1, 2, 3$) is the hopping parameter along the direction of lattice vectors δ_i , which are given by

$$\delta_1 = ae_x + \sqrt{3}ae_y, \delta_2 = ae_x - \sqrt{3}ae_y, \text{ and } \delta_3 = 2ae_x, \quad (2)$$

where e_x and e_y are basis vectors of the Cartesian coordinate system, and a is the distance between neighboring sites.

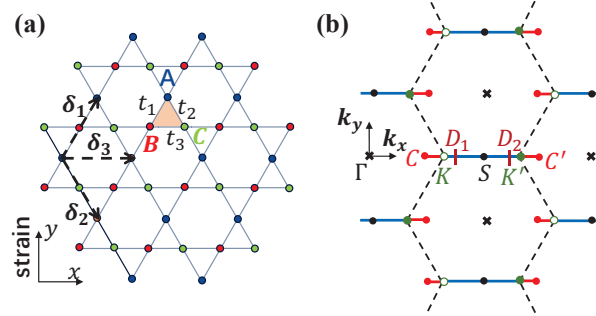


Fig. 1. (color online) (a) Tight-binding model of kagome lattice with only nearest-neighbour hopping terms. The three inequivalent sites in the unit cell are labeled as A , B and C , respectively; t_i ($i = 1, 2, 3$) denotes the hopping parameter between the nearest neighboring atoms along direction δ_i . A uniaxial strain in y direction is modeled by a group of anisotropic hopping parameters $t_1 = t_2 = t'$ and $t_3 = t$. (b) The repeated Brillouin zone of kagome lattice. The center point Γ and two independent corners K and K' are denoted by the black crosses, hollow green circles and solid green circles, respectively. When the lattice is strained in y direction, the Dirac points D_1 and D_2 are shifted away from K and K' and located somewhere between C and C' .

By applying the Wannier transformation

$$\begin{pmatrix} a_{\mathbf{r}} \\ b_{\mathbf{r}} \\ c_{\mathbf{r}} \end{pmatrix} = \sum_{\mathbf{k} \in \text{BZ}} \exp(i\mathbf{k} \cdot \mathbf{r}) \begin{pmatrix} a_{\mathbf{k}} \\ b_{\mathbf{k}} \\ c_{\mathbf{k}} \end{pmatrix} \quad (3)$$

to Eq.(1), the Hamiltonian can be rewritten in momentum space as $H = \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger \mathcal{H}_{\mathbf{k}} \Psi_{\mathbf{k}}$, where $\Psi_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, c_{\mathbf{k}})^T$ and the Hamiltonian matrix $\mathcal{H}_{\mathbf{k}}$ reads

$$\mathcal{H}_{\mathbf{k}} = \begin{pmatrix} 0 & -t_1(1 + e^{ik_1}) & -t_2(1 + e^{-ik_2}) \\ -t_1(1 + e^{-ik_1}) & 0 & -t_3(1 + e^{-ik_3}) \\ -t_2(1 + e^{ik_2}) & -t_3(1 + e^{ik_3}) & 0 \end{pmatrix}. \quad (4)$$

Here $k_i \equiv \mathbf{k} \cdot \boldsymbol{\delta}_i$ ($i = 1, 2, 3$), which satisfies $k_1 + k_2 = k_3$.

In an unstrained kagome lattice, the 6-fold rotational symmetry ensures $t_1 = t_2 = t_3$. However, as shown in Fig. 1(a), an uniaxial strain applied in y direction breaks the rotational symmetry and consequently leads to $t_1 = t_2 \neq t_3$. Assuming that $t_1 = t_2 = t'$ and $t_3 = t$, the anisotropy ratio α can be defined as $\alpha \equiv t'/t$. The parameter α characterizes both the type and strength of the uniaxial strain, as explained as follows: (i) A decrease in inter-site distance always causes an increment of hopping parameter and *vice versa*, and hence $\alpha > 1$ ($\alpha < 1$) corresponds to the situation where a compressive strain (stretching strain) is applied along y direction. (ii) Apparently, the further α deviates from 1, the stronger the applied strain is.

Meanwhile, as a reasonable approximation for weak strain[23], we neglect the site shift in the strained lattice and assume that the lattice will keep its shape when $\alpha \neq 1$. Based on this assumption, a uniaxial strain applied in y direction can be modeled using a single parameter α , which greatly reduces the mathematical complexity. The eigenvalue equation of $\mathcal{H}_{\mathbf{k}}$ is consequently reduced to

$$E_{\mathbf{k}}^3 + 4\alpha^2 t^3 (1 + \sum_i \cos k_i) - 2E_{\mathbf{k}} t^2 (\cos k_1 + 1) - 2E_{\mathbf{k}} t^2 \alpha^2 (\cos k_2 + \cos k_3 + 2) = 0. \quad (5)$$

B. Results and discussions

1. General features of energy bands

The energy bands of strained kagome lattice can be obtained by solving Eq.(5). Figs. 2(a)-(c) shows the band structures calculated with different values of α .

In the unstrained case where $\alpha = 1.0$, the solutions of Eq.(5) gives two dispersive bands $E_{\mathbf{k}}^{(1,2)} = t[-1 \pm \sqrt{2 \sum_i \cos k_i + 3}]$, and a 2D flat band $E_{\mathbf{k}}^{(3)} = 2t$. In resemblance to the energy bands of graphene, the two lower dispersive bands $E_{\mathbf{k}}^{(1)}$ and $E_{\mathbf{k}}^{(2)}$ intersect at the corners of the hexagonal Brillouin zone, forming two linearly dispersive Dirac cones at K and K' , as shown in Fig. 2(a). The flat band $E_{\mathbf{k}}^{(3)}$, which originates from the destructive interference of subdimensional eigen-wavefunctions due to the lattice geometry[13, 34, 39], touches the second band $E_{\mathbf{k}}^{(2)}$ at the center of the Brillouin zone (the Γ point).

The band structures for $\alpha = 0.8$ and $\alpha = 1.2$ were obtained by solving Eq.(5) numerically, as presented in Figs. 1(b)-(c). Similar to the situation in graphene[23], when a uniaxial strain is applied in y direction, the Dirac points of kagome lattice are shifted horizontally away from K and K' . Besides, the applied strain deforms the flat band $E_{\mathbf{k}}^{(3)}$, resulting in a band crossing between $E_{\mathbf{k}}^{(2)}$ and $E_{\mathbf{k}}^{(3)}$.

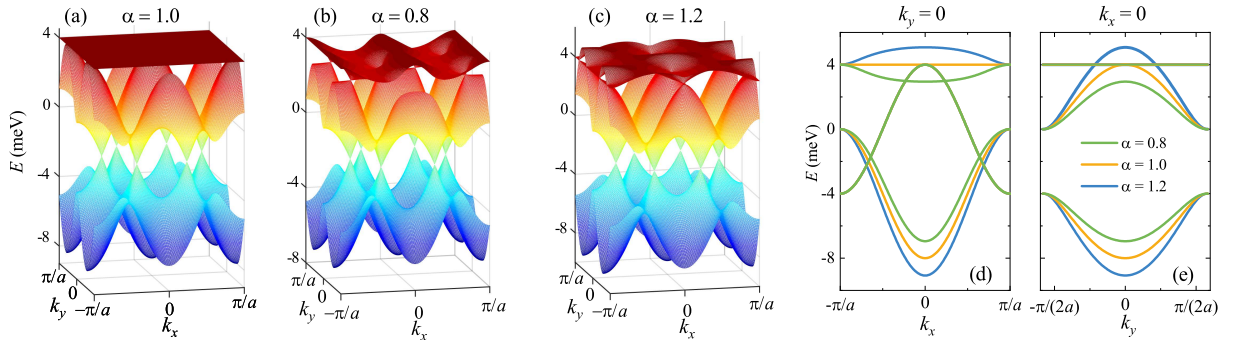


Fig. 2. (Color online) (a)-(c) Band structures of strained kagome lattices with various anisotropy ratio $\alpha \equiv t'/t$, calculated using a tight-binding model with only nearest-neighbour hopping. The lattice is (a) unstrained ($\alpha = 1.0$), (b) stretched ($\alpha = 0.8$), and (c) compressed ($\alpha = 1.2$) along y direction. (d)-(e) Band structures plotted along (d) the k_x axis and (e) the k_y axis. The hopping parameter along x direction is set to $t = t_3 = 2$ meV.

2. Shift of Dirac points

The shift of Dirac points are more visible in the energy bands plotted along the k_x axis. As shown in Fig. 2(d), the intersection points of the two lower bands, i.e., the Dirac points, move with applied strain. The horizontal position